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Arthur Cayley

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**681 (continued):** ON THE DERIVATIVES OF THREE BINARY QUANTICS.

and we obtain in like manner the equations

$$\begin{aligned}
\lambda + \lambda' &= \{\alpha + \beta 234\}\{\beta 01\}, \\
\lambda' + \lambda'' &= \{, \}\{\alpha 0\}\{\beta 0\}, \\
\lambda'' + \lambda''' &= \{, \}\{\alpha 01\}, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, +3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \lambda, \lambda', \lambda'', \lambda''') &= 0; \\
\mu + \mu' &= \{\alpha + \beta 134\} \cdot 2\{\beta 1\}, \\
\mu' + \mu'' &= \{, \} \cdot \alpha - \beta, \\
\mu'' + \mu''' &= \{, \} \cdot -2\{\alpha 1\}, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, +3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \mu, \mu', \mu'', \mu''') &= 0; \\
\nu + \nu' &= \{\alpha + \beta 034\} \cdot 1, \\
\nu' + \nu'' &= \{, \} \cdot -1, \\
\nu'' + \nu''' &= \{, \} \cdot 1, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \nu, \nu', \nu'', \nu''') &= 0; \\
\rho + \rho' &= 0, \quad \rho' + \rho'' = 0, \quad \rho'' + \rho''' = 0, \\
(\{\alpha 012\}, 3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \rho, \rho', \rho'', \rho''') &= \{\alpha + \beta 01234\}.
\end{aligned}$$

These give for the  $\psi'''$  square the values

$$\begin{aligned}
&\{\beta 012\}, \quad 3\{\beta 12\}, \quad 3\{\beta 2\}, \quad +\dots, \quad 1, \\
&\{\alpha 0\}\{\beta 01\}, \quad -2\alpha\beta \cdot \{\beta 1\}, \quad -\alpha^2\beta^2, \quad 1, \\
&\{\alpha 01\}\{\beta 0\}, \quad +\alpha^2\beta \cdot \{\alpha 1\}, \quad 2\alpha\beta^2, \quad 1, \\
&-\{\alpha 012\}, \quad 3\{\alpha 12\}, \quad 3\{\alpha 2\}, \quad 1,
\end{aligned}$$

and so on; the law however of the terms in the intermediate lines is not by any means obvious.

Consider now the binary quantics  $P, Q, R$ , of the forms  $(* \$ x, y)^p, (* \$ x, y)^q, (* \$ x, y)^r$ ; we have for any, for instance for the fourth, order, the derivates

$$P(Q, R)^4, (P, (Q, R)^3)^1, (P, (Q, R)^2)^2, (P, (Q, R)^1)^3, (P, QR)^4;$$

and it is required to express

$$Q(P, R)^4 \text{ and } R(P, Q)^4,$$

each of them as a linear function of these.

I recall that we have  $(P, Q)^0 = PQ$ , so that the first and the last terms of the series might have been written  $(P, (Q, R)^4)^0$  and  $(P, (Q, R)^0)^4$  respectively; and, further, that  $(P, Q)^2$  denotes  $d_x P \cdot d_y Q - d_y P \cdot d_x Q$ ;  $(P, Q)^2$  denotes

$$d_x^2 P \cdot d_y^2 Q - 2d_x d_y P \cdot d_x d_y Q + d_y^2 P \cdot d_x^2 Q;$$

and so on.

I write  $(a, b, c, d, e)$  for the fourth derived functions of any quantic  $U, = (*\$x, y)^m$ ; we have, in a notation which will be at once understood,

$$\begin{aligned} U &= (a, b, c, d, e \$x, y)^4 \div [m]^4, \\ (d_x, d_y) U &= ((a, b, c, d), (b, c, d, e) \$x, y)^3 \div [m-1]^3, \\ (d_x, d_y)^2 U &= ((a, b, c), (b, c, d), (c, d, e) \$x, y)^2 \div [m-2]^2, \\ (d_x, d_y)^3 U &= ((a, b), (b, c), (c, d), (d, e) \$x, y) \div [m-3], \\ (d_x, d_y)^4 U &= (a, b, c, d, e); \end{aligned}$$

and then, taking

$$(a_1, b_1, c_1, d_1, e_1), (a_2, b_2, c_2, d_2, e_2), (a_3, b_3, c_3, d_3, e_3),$$

to belong to  $P, Q, R$ , respectively, we must, instead of  $m$ , write  $p, q, r$  for the three functions respectively.

If we attend only to the highest terms in  $x$ , we have

$$\begin{aligned} U &= ax^m + [m-1], \\ (d_x, d_y) U &= (a, b) x^{m-1} + [m-2], \\ (d_x, d_y)^2 U &= (a, b, c) x^{m-2} + [m-3], \\ (d_x, d_y)^3 U &= (a, b, c, d) x^{m-3} + [m-4], \\ (d_x, d_y)^4 U &= (a, b, c, d, e). \end{aligned}$$

Consider now  $P(Q, R)^4, (P, (Q, R)^3)^1$ , etc.; in each case attending only to the term in  $a_1$ , and in this term to the highest term in  $x$ , we have

- (1)  $[p]^0 P(Q, R)^4 = a_1 \{a_3 e_3 - 4b_3 d_3 + 6c_2 c_3 - 4d_2 b_3 + e_2 a_3\} \quad (X),$
- (2)  $[p-1]^1 [q-3]^3 [r-3]^3 (P, (Q, R)^3)^1 = -[q-3]^3 \cdot b_1 \{d_2 \cdot 3c_2 c_3 - 3d_2 b_3 - e_2 a_3\} \quad (-Y),$   
 $+ [r-3]^3 \cdot a_1 a_2 e_3 - 3b_2 d_3 + 3c_2 c_3 - d_2 b_3 \quad (Y),$
- (3)  $[p-2]^2 [q-2]^2 [r-2]^2 (P, (Q, R)^2)^2 = [q-2]^2 c_1 c_2 a_3 - 2d_2 b_3 + e_2 a_3 \quad (Z''),$   
 $+ 2[q-2]^2 [r-2]^2 \cdot b_1 d_2 - 2c_2 c_3 + d_2 b_3 \quad (-Z'),$   
 $+ [r-2]^2 c_1 e_3 - 2b_2 d_3 + c_2 c_3 \quad (Z),$
- (4)  $[p-3]^3 [q-1]^1 [r-1]^1 (P, (Q, R)^1)^3 = [q-1]^1 d_1 b_3 - c_2 a_3 \quad (-W''),$   
 $+ 3[q-1]^1 [r-1]^1 c_1 c_3 - d_2 b_3 \quad (W''),$   
 $+ 3[q-1]^1 [r-1]^1 b_1 d_3 - c_2 c_3 \quad (-W'),$   
 $+ [r-1]^1 a_1 e_3 - b_2 d_3 \quad (W),$

$$\begin{aligned}
 (5) \quad & [p-4]^4 [q]^0 [r]^0 (P, QR)^4 = e_1 a_3 a_3 (U''''), \\
 & + 4[q]^0 [r]^0 d_1 b_3 (-U'''), \\
 & + 6[q]^0 [r]^0 c_1 c_3 (U''), \\
 & + 4[q]^0 [r]^0 b_1 d_3 (-U'), \\
 & + [q]^0 [r]^0 a_1 e_3 (U).
 \end{aligned}$$

Thus, for the second of these equations,

$$(P, (Q, R)^3)^1 = d_x P \cdot d_y (Q, R)^3 - \&c.$$

the term in  $a_1$  is  $d_y(Q, R)^3 = (d_x Q, R)^3 + (Q, d_y R)^3$ , the whole being divided by  $[p-1]^1$ ; where attending only to the highest terms in  $x$ , the two terms are respectively

$$-(b_2 d_3 - 3c_2 c_3 + 3d_2 b_3 - e_2 a_3) \div [r-3]^3,$$

and

$$(a_2 e_3 - 3b_2 d_3 + 3c_2 c_3 - d_2 b_3) \div [q-3]^3,$$

which are each divided by  $[p-1]^1$  as above; whence, multiplying by

$$[p-1]^1 [q-1]^1 [r-1]^1,$$

we have the formula in question; and so for the other cases.

Writing now (1), (2), (3), (4), (5) for the left-hand sides of the five equations respectively; and

$$-Y', Y; \quad Z'', -Z', Z; \quad -W''', W'', -W', W; \quad U'''', -U''', U'', -U', U;$$

for the literal parts on the right-hand sides of the same equations respectively, then we have

$$X = Y + Y',$$

$$Y = Z + Z', \quad Y' = Z' + Z'',$$

$$\&c.,$$

and the equations become

$$(1) = X,$$

$$(2) = [r-1]^1 Y - [q-3]^3 Y',$$

$$(3) = [r-2]^2 Z - 2[r-2]^2 [q-2]^2 Z' + [q-2]^2 Z'',$$

$$(4) = [r-1]^1 W'' - 3[r-1]^1 [q-1]^1 W' + 3[r-1]^1 [q-1]^1 W - [q-1]^1 W''',$$

$$(5) = [r]^0 U - 4[r]^0 [q]^0 U' + 6[r]^0 [q]^0 U'' - 4[r]^0 [q]^0 U''' - [q]^0 U'''' ,$$

which are, in fact, the equations considered at the beginning of the present paper, putting therein  $\alpha = r-3$  and  $\beta = q-3$ , they consequently give

$$\{q+r-6, 456\}(1), \{q+r-6, 156\}(2), \{q+r-6, 036\}(3), +\{q+r-6, 015\}(4), \{q+r-6, 012\}(5),$$

$$\{q+r-6, 01\dots 6\}Q = \{q-3, 0123\} + 4\{q-3, 123\} + 6\{q-3, 23\} + 4\{q-3, 3\} + 1,$$

$$U'''' = \{r-3, 0123\} - 4\{r-3, 123\} + 6\{r-3, 23\} - 4\{r-3, 3\} + 1.$$

Also, attending as before only to the terms in  $a_1$ , and therein to the highest power of  $x$ , we have

$$Q(P, R)^4 = a_2 a_1 \div [q]^0,$$

that is,

$$[q]^2 Q(P, R)^4 = U, \quad [r]^0 R(P, Q)^4 = U'''';$$

and, observing that  $\{q + r - 6, 01 \dots 6\}$  is  $= [q + r]^6$ , and that  $\{q + r - 6, 456\}$ , etc., may be written  $\{q + r, \bar{2}\bar{1}\bar{0}\}$ , etc., where the superscript bars are the signs  $-$ , the formulae become

$$\begin{aligned} &\{q + r, \bar{2}\bar{1}\bar{0}\}(1), \quad \{q + r, \bar{5}\bar{1}\bar{0}\}(2), \quad \{q + r, \bar{6}\bar{3}\bar{0}\}(3), \quad \{q + r, \bar{6}\bar{5}\bar{1}\}(4), \quad \{q + r, \bar{6}\bar{5}\bar{4}\}(5), \\ &[q + r]^6 [q]^0 Q(P, R)^4 = [q]^0 + 6[q]^0 + 4[q]^0 + [q]^0 + 1, \\ &[q + r]^6 [r]^0 R(P, Q)^4 = [r]^0 - 4[r]^0 + 6[r]^0 - 4[r]^0 + 1. \end{aligned}$$

Written at full length, the first of these equations (which, as being the fourth in a series, I mark 4th equation) is

$$\begin{aligned} [q + r]^6 [q]^0 Q(P, R)^4 &= 1 \cdot q + r \cdot q + r - 1 \cdot q + r - 2 \cdot [p]^0 [q]^0 \cdot P(Q, R)^4 \quad (4\text{th equation}) \\ &+ 4 \cdot q + r \cdot q + r - 1 \cdot q + r - 5 \cdot [p - 1]^1 [q]^0 [q - 3]^3 [r - 1]^1 \cdot (P, (Q, R)^3)^1 \\ &+ 6 \cdot q + r \cdot q + r - 5 \cdot q + r - 6 \cdot [p - 2]^2 [q]^0 [q - 2]^2 [r - 2]^2 \cdot (P, (Q, R)^2)^2 \\ &+ 4 \cdot q + r - 1 \cdot q + r - 5 \cdot q + r - 6 \cdot [p - 3]^3 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, (Q, R)^1)^3 \\ &+ 1 \cdot q + r - 1 \cdot q + r - 5 \cdot q + r - 6 \cdot [q]^0 \cdot P(Q, R)^4 [r]^0; \end{aligned}$$

and the other is, in fact, the same equation with  $q, Q, r, R$  interchanged with  $r, R, q, Q$ ; the alternate  $+$  and  $-$  signs arise evidently from the terms

$$(R, Q)^4 = (Q, R)^4; \quad (R, Q)^3 = -(Q, R)^3; \quad \text{etc.},$$

which present themselves on the right-hand side.

It will be observed that the identity has been derived from the comparison of the terms in  $a_1$ , which are the highest terms in  $x$ , the other terms not having been written down or considered; but it is easy to see that an identity of the form in question exists, and, this being admitted, the process is a legitimate one.

The preceding equations of the series are

$$\begin{aligned} [q + r]^2 [q]^0 Q(P, R)^2 &= 1 \cdot [p]^0 [q]^0 \cdot P(Q, R)^2 \quad (1\text{st equation}) \\ &+ 1 \cdot [q]^0 [r]^0 \cdot (P, QR)^2; \\ [q + r]^3 [q]^0 Q(P, R)^3 &= 1 \cdot q + r \cdot [p]^0 [q]^0 \cdot P(Q, R)^3 \quad (2\text{nd equation}) \\ &+ 2 \cdot q + r - 1 \cdot [p - 1]^1 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, (Q, R)^1)^1 \\ &+ 1 \cdot q + r - 2 \cdot [q]^0 [r]^0 \cdot (P, QR)^3; \\ [q + r]^4 [q]^0 Q(P, R)^4 &= 1 \cdot q + r \cdot q + r - 1 \cdot [p]^0 [q]^0 \cdot P(Q, R)^4 \quad (3\text{rd equation}) \\ &+ 3 \cdot q + r \cdot q + r - 3 \cdot [p - 1]^1 [q]^0 [q - 2]^2 [r - 2]^2 \cdot (P, (Q, R)^2)^1 \\ &+ 2 \cdot q + r - 1 \cdot q + r - 4 \cdot [p - 2]^2 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, QR)^2 \\ &+ 1 \cdot q + r - 3 \cdot q + r - 4 \cdot [q]^0 [r]^0 \cdot (P, QR)^4. \end{aligned}$$

From these four equations the law is evident, except as to the numbers subtracted from  $q + r$ . These are obtained, as explained above, in regard to the numbers added to  $\alpha + \beta$  in the  $\{ \}$  symbols; transforming the diagrams so as to be directly applicable to the case now in question, they become

$$\begin{array}{cccccc}
 01 & 01 & 012 & 012 & 0123 & 0123 \\
 1\ 1 & 2\ 03 & 3\ 015 & 4\ 0127 & & \\
 1\ 2 & 1\ 1\ 14 & 21\ 036 & 31\ 0158 & & \\
 & 2\ 34\ 12\ 156 & & 22\ 0378 & & \\
 & & 3\ 456\ 13 & & 1678 & \\
 & & & 4\ 5678 & & 
 \end{array}$$

showing how the numbers are obtained for the equations 2, 3, 4, 5 respectively.

The first equation is

$$(q^2 + qr) Q(P, R) = pq P(Q, R) + qr [Q(P, R) + (P, Q)],$$

viz. this is

$$(q^2 + qr) Q(P, R) = pq P(Q, R) - qr Q(R, P) + qr R(P, Q)$$

or, dividing by  $q$ , this is

$$(q^2 + qr) Q(R, P) = p P(Q, R) + q Q(R, P) + r R(P, Q),$$

which is a well-known identity.

We may verify any of the equations, though the process is rather laborious, for the particular values

$$P = Q = R = x^{\frac{1}{2}(p+\alpha)} y^{\frac{1}{2}(p-\alpha)}, \quad x^{\frac{1}{2}(q+\beta)} y^{\frac{1}{2}(q-\beta)}, \quad x^{\frac{1}{2}(r+\gamma)} y^{\frac{1}{2}(r-\gamma)}.$$

Thus, taking the second equation, we have, omitting common factors,

$$\begin{aligned}
 (Q, R)^2 &= q + \beta \cdot q + \beta - 2 \cdot r - \gamma \cdot r - \gamma - 2 \\
 &\quad - 2 \cdot q + \beta \cdot q - \beta \cdot r + \gamma \cdot r - \gamma \\
 &\quad + q - \beta \cdot q - \beta - 2 \cdot r + \gamma \cdot r + \gamma - 2 \\
 &= \beta^2(r^2 - r) + \gamma^2(q^2 - q) - 2\beta\gamma(q - 1)(r - 1) - qr(q + r - 2),
 \end{aligned}$$

$$\begin{aligned}
 (P, (Q, R)^1)^1 &= (q + \beta \cdot r - \gamma - q - \beta \cdot r + \gamma)(p + \alpha \cdot q + r - \beta - \gamma - 2 - p - \alpha \cdot q + r + \beta + \gamma - 2) \\
 &= (\beta r - q\gamma)(\alpha \cdot q + r - 2 - p \cdot \beta + \gamma) \\
 &= \alpha\beta r(r + q - 2) - \alpha\gamma q(q + r - 2) - pr\beta^2 + p(q - r)\beta\gamma + pq\gamma^2,
 \end{aligned}$$

and from the first of these the expressions of  $Q(P, R)^2$  and  $(P, QR)^2$  are at once obtained. The identity to be verified then becomes

$$\begin{aligned}
 &[q + r]^2 [q]^2 \{ \alpha^2(r^2 - r) + \gamma^2(p^2 - p) - 2\alpha\gamma(p - 1)(r - 1) - pr(p + r - 2) \} \\
 &= (q + r)[q]^2 [p]^2 \{ \beta^2(r^2 - r) + \gamma^2(q^2 - q) - 2\beta\gamma(q - 1)(r - 1) - qr(q + r - 2) \} \\
 &\quad + 2(q + r - 1)[q]^2 (p - 1)(r - 1) \{ \alpha\beta r(q + r - 2) - \alpha\gamma q(q + r - 2) \\
 &\quad \quad - pr\beta^2 + p(q - r)\beta\gamma + pq\gamma^2 \} \\
 &\quad + (q + r - 2)[q]^2 [r]^2 \{ \alpha^2(q + r)(q + r - 1) + (\beta + \gamma)^2(p^2 - p) \\
 &\quad \quad - 2\alpha(\beta + \gamma)(p - 1)(q + r - 1) - p(q + r)(p + q + r - 2) \},
 \end{aligned}$$

which is easily verified, term by term; for instance, the terms with  $\alpha$ ,  $\beta$ , or  $\gamma$ , give

$$\begin{aligned} [q+r]^2[q]^2 pr(p+r-2) &= (q+r)[q]^2[p]^2 qr(q+r-2) \\ &\quad + (q+r-2)[q]^2[r]^2 p(q+r)(p+q+r-2), \end{aligned}$$

which, omitting the factor  $(q+r)(q+r-2)[q]^2 pr$ , is

$$(q+r-1)(p+r-2) = (p-1)q + (r-1)(p+q+r+2);$$

viz. the right-hand side is

$$(p-1)q + (r-1)q + (r-1)(p+r-2) = (q+r-1)(p+r-2),$$

as it should be.

The equations are useful for the demonstration of a subsidiary theorem employed in Gordan's demonstration of the finite number of the covariants of any binary form  $U$ . Suppose that a system of covariants (including the quantic itself) is

$$P, Q, R, S, \dots;$$

this may be the complete system of covariants; and if it is so, then,  $T$  and  $V$  being any functions of the form  $P^\alpha Q^\beta R^\gamma \dots$ , every derivative  $(T, V)^s$  must be a term or sum of terms of the like form  $P^\alpha Q^\beta R^\gamma \dots$ ; the subsidiary theorem is that in order to prove that the case is so, it is sufficient to prove that every derivative  $(P, Q)^s$ , where  $P$  and  $Q$  are any two terms of the proposed system, is a term or sum of terms of the form in question  $P^\alpha Q^\beta R^\gamma \dots$ .

In fact, supposing it shown that every derivative  $(T, V)^s$  up to a given value  $\theta$  of  $s$  is of the form  $P^\alpha Q^\beta R^\gamma \dots$ , we can by successive application of the equation for  $Q(P, R)^{\theta+1}$ , regarded as an equation for the reduction of the last term on the right-hand side  $(P, QR)^{\theta+1}$ , bring first  $(P, QR)^{\theta+1}$ , and then  $(P, QRS)^{\theta+1}, \dots$ , and so ultimately any function  $(P, V)^{\theta+1}$ , and then again any functions  $(PQ, V)^{\theta+1}, (PQR, V)^{\theta+1}, \dots$ , and so ultimately any function  $(T, V)^{\theta+1}$ , into the required form  $P^\alpha Q^\beta R^\gamma \dots$ : or the theorem, being true for  $\theta$ , will be true for  $\theta + 1$ ; whence it is true generally.

## 682.

## FORMULÆ RELATING TO THE RIGHT LINE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 169–171.]

1. Let  $\lambda, \mu, \nu$  be the direction-angles of a line;  $\alpha, \beta, \gamma$  the coordinates of a point on the line; and write

$$\begin{aligned} a &= \cos \lambda, & f &= \beta \cos \nu - \gamma \cos \mu, \\ b &= \cos \mu, & g &= \gamma \cos \lambda - \alpha \cos \nu, \\ c &= \cos \nu, & h &= \alpha \cos \mu - \beta \cos \lambda, \end{aligned}$$

whence

$$a^2 + b^2 + c^2 = 1,$$

$$af + bg + ch = 0,$$

or the six quantities  $(a, b, c, f, g, h)$ , termed the coordinates of the line, depend upon four arbitrary parameters.

2. It is at once shown that the condition for the intersection of any two lines  $(a, b, c, f, g, h)$ ,  $(a', b', c', f', g', h')$ , is  $af' + bg' + ch' + a'f + b'g + c'h = 0$ .

3. Given two lines  $(a, b, c, f, g, h)$ ,  $(a', b', c', f', g', h')$ , it is required to find their shortest distance, and the coordinates of their line of shortest distance.

Let

$$Ax + By + Cz + D = 0,$$

$$Ax + By + Cz + D' = 0,$$

be parallel planes containing the two lines respectively; then the first plane contains the point  $\alpha + r \cos \lambda$ ,  $\beta + r \cos \mu$ ,  $\gamma + r \cos \nu$ , and the second contains the point  $\alpha' + r' \cos \lambda'$ ,  $\beta' + r' \cos \mu'$ ,  $\gamma' + r' \cos \nu'$ ; that is, we have

$$A\alpha + B\beta + C\gamma + D = 0,$$

$$A\alpha' + B\beta' + C\gamma' + D' = 0,$$

$$A \cos \lambda + B \cos \mu + C \cos \nu = 0,$$

$$A \cos \lambda' + B \cos \mu' + C \cos \nu' = 0,$$

which last equations may be written

$$\begin{aligned} Aa + Bb + Cc &= 0, \\ Aa' + Bb' + Cc' &= 0, \end{aligned}$$

giving

$$A : B : C = bc' - b'c : ca' - c'a : ab' - a'b,$$

or, if we write

$$\theta = aa' + bb' + cc',$$

and assume, as is convenient,

$$A^2 + B^2 + C^2 = 1,$$

then

$$\sqrt{1 - \theta^2} A, B, C = bc' - b'c, ca' - c'a, ab' - a'b,$$

where  $\theta = \text{cosine-inclination} = aa' + bb' + cc'$ .

Hence, shortest distance  $= D' - D$

$$\begin{aligned} &= A(\alpha - \alpha') + B(\beta - \beta') + C(\gamma - \gamma') \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ (bc' - b'c)(\alpha - \alpha') + (ca' - c'a)(\beta - \beta') + (ab' - a'b)(\gamma - \gamma') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ a'(c\beta - b\gamma) + b'(a\gamma - c\alpha) + c'(b\alpha - a\beta) \\ &\quad + a(c'\beta' - b'\gamma') + b(a'\gamma' - c'\alpha') + c(b'\alpha' - a'\beta') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} (af' + bg' + ch' + a'f + b'g + c'h), = \mathfrak{D} \text{ suppose.} \end{aligned}$$

The six coordinates of the line of shortest distance are  $A, B, C, F, G, H$ , where  $A, B, C$  denote as before, and  $F, G, H$  are to be determined.

Since the line meets each of the given lines, we have

$$\begin{aligned} Af + Bg + Ch + Fa + Gb + Hc &= 0, \\ Af' + Bg' + Ch' + Fa' + Gb' + Hc' &= 0, \\ FA + GB + HC &= 0, \end{aligned}$$

which equations give  $F, G, H$ . Multiplying the first equation by  $-b'C + c'B$ , the second by  $Bc - Cb$ , and the third by  $-bc' + b'c$ , we find

$$(b'C - c'B)(Af + Bg + Ch) + (Bc - Cb)(Af' + Bg' + Ch') + F \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ A, & B, & C \end{vmatrix} = 0.$$

Here

$$\begin{aligned} b'C - c'B &= \frac{1}{\sqrt{1 - \theta^2}} \{ b'(ab' - a'b) - c'(ca' - c'a) \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ a(a'^2 + b'^2 + c'^2) - a'(aa' + bb' + cc') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} (a - a'\theta), \end{aligned}$$



and similarly

$$cB - bC = \frac{1}{\sqrt{1-\theta^2}}(a' - a\theta).$$

Also, putting for shortness

$$\Omega = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f, & g, & h \end{vmatrix}, \quad \Omega' = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f', & g', & h' \end{vmatrix},$$

we have

$$Af + Bg + Ch = \frac{1}{\sqrt{1-\theta^2}} \Omega, \quad Af' + Bg' + Ch' = \frac{1}{\sqrt{1-\theta^2}} \Omega',$$

and finally, the determinant which multiplies  $F$  is

$$\frac{1}{\sqrt{1-\theta^2}} \{(bc' - b'c)^2 + (ca' - c'a)^2 + (ab' - a'b)^2\} = \frac{1}{\sqrt{1-\theta^2}}(1 - \theta^2), = \sqrt{1-\theta^2}.$$

We have thus the value of  $F$ ; forming in the same way those of  $G$  and  $H$ , we find

$$\begin{aligned} F &= \frac{-1}{(1-\theta^2)^{\frac{3}{2}}} \{(a - a'\theta) \Omega + (a' - a\theta) \Omega'\}, \\ G &= \frac{-1}{(1-\theta^2)^{\frac{3}{2}}} \{(b - b'\theta) \Omega + (b' - b\theta) \Omega'\}, \\ H &= \frac{-1}{(1-\theta^2)^{\frac{3}{2}}} \{(c - c'\theta) \Omega + (c' - c\theta) \Omega'\}, \end{aligned}$$

which, with the foregoing equations for  $A, B, C$ , give the six coordinates of the line of shortest distance.

## 683.

ON THE FUNCTION  $\arcsin(x + iy)$ .

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 171–174.]

The determination of the function in question, the arc to a given imaginary sine, is considered in Cauchy's *Exercices d'Analyse, &c.*, t. iii. (1844), p. 382; but it appears, by two hydrodynamical papers by Mr Ferrers and Mr Lamb, *Quarterly Mathematical Journal*, t. XIII. (1874), p. 115, and t. XIV. (1875), p. 40, that the question is connected with the theory of confocal conics.

Taking  $c = \sqrt{a^2 - b^2}$  a positive real quantity which may ultimately be put  $= 1$ , the question is to find the real quantities  $\xi$ ,  $\eta$ , such that

$$\xi + i\eta = \arcsin \frac{1}{c}(x + iy),$$

or say

$$x + iy = c \sin(\xi + i\eta),$$

so that

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta.$$

It is convenient to remark that if a value of  $\xi + i\eta$  be  $\xi' + i\eta'$ , then the general value is  $2m\pi + \xi' + i\eta'$  or  $(2m + 1)\pi - (\xi' + i\eta')$ ; hence,  $\eta$  may be made positive or negative at pleasure;  $\cos i\eta$  is in each case positive, but  $-\sin i\eta$  has the same sign as  $\eta$ ; hence  $\cos \xi$  has the same sign as  $x$ , but  $\sin \xi$  has the same sign as  $y$  or the reverse sign, according as  $\eta$  is positive or negative; for any given values of  $x$  and  $y$ , we obtain, as will appear, determinate positive values of  $\sin^2 \xi$  and  $\cos^2 \xi$ ; and the square roots of these must therefore be taken so as to give to  $\sin \xi$ ,  $\cos \xi$  their proper signs respectively.

Suppose that  $\lambda, \mu$  are the elliptic coordinates of the point  $(x, y)$ ; viz. that we have

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1,$$

$$\frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1,$$

where  $a^2 + \lambda, b^2 + \lambda$ , and  $a^2 + \mu$  are positive, but  $b^2 + \mu$  is negative. Calling  $\rho, \sigma$  the distances of the point  $x, y$  from the points  $(c, 0)$  and  $(-c, 0)$ , that is, assuming

$$\rho = \sqrt{(x - c)^2 + y^2},$$

$$\sigma = \sqrt{(x + c)^2 + y^2},$$

then we have

$$\begin{aligned} \sqrt{a^2 + \lambda} &= \frac{1}{2}(\sigma + \rho), & \text{whence also } \sqrt{b^2 + \lambda} &= \frac{1}{2}\sqrt{(\sigma + \rho)^2 - 4c^2}, \\ \sqrt{a^2 + \mu} &= \frac{1}{2}(\sigma - \rho), & \sqrt{b^2 + \mu} &= \frac{1}{2}\sqrt{(\sigma - \rho)^2 - 4c^2}, \end{aligned}$$

which equations determine  $\lambda, \mu$  as functions of  $x, y$ .

Now we have

$$\begin{aligned} \rho\sigma &= \sqrt{(x^2 + y^2 - c^2)^2 - 4c^2y^2}, \\ \rho^2 + \sigma^2 &= 2(x^2 + y^2 + c^2); \end{aligned}$$

substituting herein for  $x, y$  their values

$$c \sin \xi \cos i\eta, \quad ci \cos \xi \sin i\eta,$$

we find

$$\begin{aligned} x^2 - y^2 - c^2 &= c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta - \sin^2 \xi - \cos^2 \xi) \\ &= -c^2(\sin^2 \xi \sin^2 i\eta + \cos^2 \xi \cos^2 i\eta), \end{aligned}$$

whence

$$\begin{aligned} (x^2 - y^2 - c^2)^2 &= c^4(\cos^2 \xi \cos^2 i\eta + \sin^2 \xi \sin^2 i\eta)^2, \\ 4x^2y^2 &= 4c^4 \sin^2 \xi \cos^2 \xi \sin^2 i\eta \cos^2 i\eta \\ &= c^4(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta)^2. \end{aligned}$$

Hence  $\rho\sigma = 2c^2(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta)$ , and

$$\begin{aligned} \rho^2 + \sigma^2 &= 2c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta + 1) \\ (\rho + \sigma)^2 &= 2c^2(\cos^2 i\eta + \sin^2 i\eta + 1) = 4c^2 \cos^2 i\eta, \\ (\rho - \sigma)^2 &= 2c^2(\sin^2 \xi - \cos^2 \xi + 1) = 4c^2 \sin^2 \xi. \end{aligned}$$

Consequently

$$\begin{aligned} a^2 + \lambda &= c^2 \cos^2 i\eta, & b^2 + \lambda &= -c^2 \sin^2 i\eta, \\ a^2 + \mu &= c^2 \sin^2 \xi, & b^2 + \mu &= -c^2 \cos^2 \xi, \end{aligned}$$

values which verify as they should do the equations

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1,$$

$$\frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1.$$

viz. these become

$$\frac{x^2}{c^2 \cos^2 i\eta} + \frac{y^2}{-c^2 \sin^2 i\eta} = \sin^2 \xi + \cos^2 \xi = 1,$$

$$\frac{x^2}{c^2 \sin^2 \xi} + \frac{y^2}{-c^2 \cos^2 \xi} = \cos^2 i\eta - \sin^2 i\eta = 1.$$

The same equations, or as we may also write them,

$$\lambda = -a^2 \sin^2 i\eta - b^2 \cos^2 i\eta,$$

$$\mu = -a^2 \cos^2 \xi - b^2 \sin^2 \xi,$$

determine  $\eta$  as a function of  $\lambda$ , and  $\xi$  as a function of  $\mu$ ;  $\lambda, \mu$  being by what precedes, given functions of  $x, y$ .

Or more simply, starting from the last-mentioned values of  $\lambda, \mu$ , and substituting these in the expressions

$$x^2 = \frac{a^2 + \lambda}{a^2 - b^2} \cdot \frac{a^2 + \mu}{1}, \quad y^2 = \frac{b^2 + \lambda}{b^2 - a^2} \cdot \frac{b^2 + \mu}{1},$$

we find

$$x^2 = c^2 \sin^2 \xi \cos^2 i\eta, \quad y^2 = -c^2 \cos^2 \xi \sin^2 i\eta,$$

or say

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta,$$

whence

$$x + iy = c \sin(\xi + i\eta),$$

the original relation between  $x, y$  and  $\xi, \eta$ .

684.

ON A RELATION BETWEEN CERTAIN PRODUCTS OF  
DIFFERENCES.[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 174, 175.]

Consider the function

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\}$$

where

$$abc = (a-b)(b-c)(c-a),$$

$$ab = (a-b)(b-a) = -(a-b)^2,$$

therefore

$$abc = bca = cab = -bac, \&c.,$$

$$ab = ba.$$

It is to be shown that the function vanishes if  $e = d$ . Writing  $e = d$ , the value is

$$\begin{aligned} & 3(bcd \cdot da + dab \cdot cd) - abd \cdot cd \\ & \quad - bcd \cdot da \\ & \quad - cda \cdot db \\ & \quad - dac \cdot bd, \end{aligned}$$

viz. this is

$$\begin{aligned}
 & -3bcd \cdot ad - abd \cdot cd \\
 & +3abd \cdot cd - bcd \cdot ad \\
 & \quad - 2acd \cdot bd \\
 & = 2bcd \cdot ad - 2acd \cdot bd + 2abd \cdot cd \\
 & = 2(bcd \cdot ad + cad \cdot bd + abd \cdot cd),
 \end{aligned}$$

which is easily seen to vanish; the value is

$$\begin{aligned}
 & (b-c)(c-d)(d-b)(a-d)^2 - (b-c)(a-d)^2(b-d)(c-d) \\
 & + (c-a)(a-d)(d-c)(b-d)^2 - (c-a)(a-d)(b-d)^2(c-d) \\
 & + (a-b)(b-d)(d-a)(c-d)^2 - (a-b)(a-d)(b-d)(c-d)^2.
 \end{aligned}$$

viz. omitting the factor  $(a-d)(b-d)(c-d)$ , this is

$$-(b-c)(a-d) - (c-a)(b-d) - (a-b)(c-d),$$

which vanishes. Hence the function also vanishes if  $e = a$ , or  $a = b$  or  $b = c$ , or  $c = d$ ; and it is thus a mere numerical multiple of  $(a-b)(b-c)(c-d)(d-e)(e-a)$ , or say it is  $= M \cdot abcde$ .

To find  $M$  write  $e = c$ , the equation becomes

$$\begin{aligned}
 3abc \cdot dc - cda \cdot cb &= M \cdot abcde, = M \cdot abc \cdot dc, \\
 & +3bcd \cdot ca - ac \\
 & +3dea \cdot bc \\
 & +3cab \cdot cd,
 \end{aligned}$$

viz. this is  $6abc \cdot dc + 4dbc \cdot ac + 4cadc \cdot bc = M \cdot abc \cdot dc$ ,

giving  $M = 10$ . In fact, we then have

$$4abc \cdot dc + 4dbc \cdot ac - 4adc \cdot bc = 0,$$

that is,

$$-abc \cdot dc - bdc \cdot ac - dac \cdot bc = 0,$$

which is right. And we have thus the identity

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\} = 10 \cdot abcde,$$

or say

$$3[abcde] - [acebd] = 10\{abcde\}.$$

## 685.

## ON MR COTTERILL'S GONIOMETRICAL PROBLEM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 196–198.]

The very remarkable formulæ contained in Mr Cotterill's paper, "A goniometrical problem, to be solved analytically in one move, or more simply synthetically in two moves," *Quarterly Mathematical Journal*, t. vii. (1866), pp. 259–272, are presented in a form which, to say the least, is not as easily intelligible as might be; and they have not, I think, attracted the attention which they well deserve.

Using his notation, except that I write for angles small roman letters, in order to be able to have the corresponding italic small letters and capitals for the sines and cosines respectively of the same angles, we consider nine angles

$$\begin{array}{ccc} a, & b, & c, \\ d, & e, & f, \\ x, & y, & z, \end{array}$$

which are such that the sum of three angles in the same line, or in the same column, is an odd multiple of  $\pi$ . Of course, any four angles such as  $a, b, d, e$  are

arbitrary, and each of the remaining angles is then determinate save as to an even multiple of  $\pi$ . And it may be remarked that these angles  $a, b, d, e$  may represent the inclinations of any four lines to a fifth line, and that the remaining angles are then at once obtained, as in the figure. The small roman letters are here used to denote as well angles as points, being so placed as to show what the angles are which they respectively denote; the points  $*$ ,  $*$  are constructed as the intersections of the lines  $ac, bc$  by the circle circumscribed about  $axy$ , and the angle  $z$  is the angle which the points  $*$ ,  $*$  subtend at  $x$  or  $y$ . It will be observed that the sum of the three angles in a line or column is in each case  $= \pi$ .

But this in passing: the analytical theorem is, first, we can form with the sines and cosines of the angles in any two lines or columns a function  $S$  presenting itself under two distinct forms, which are in fact equal in value, or say  $S$  is a symmetrical function of the two lines or columns, viz. for the first and second lines this is

$$\begin{aligned} S\left(\begin{matrix} a, & b, & c \\ d, & e, & f \end{matrix}\right) &= d^2 Abc + e^2 Bca + f^2 Cab \\ &= a^2 Def + b^2 Efd + c^2 Fde, \end{aligned}$$

where, as already mentioned,  $a, A$  denote  $\sin a, \cos a$ , and so for the other letters.

Secondly, if to the  $S$  of any two lines or columns we add twice the product of the six sines, we obtain a sum  $M$  which has the same value from whichever two lines or columns we obtain it; or, say  $M$  is a symmetrical function of the matrix of the nine angles. Thus

$$M = S\left(\begin{matrix} a, & b, & c \\ d, & e, & f \end{matrix}\right) + 2abcdef,$$

which is one of a system of six forms each of which (on account of the two forms of the  $S$  contained in it) may be regarded as a double form, and the twelve values are all of them equal. There are, moreover, 15 other forms of  $M$ , viz. 3 line-forms, such as

$$bcdx + caey + abfz \quad (\text{belongs to line } a, b, c),$$

3 column-forms, such as

$$dxbc + xaef + adyz \quad (\text{belongs to column } a, d, x),$$

and 9 term-forms, such as

$$e^2 z^2 + f^2 y^2 + 2efyzA \quad (\text{belongs to term } a),$$

and the  $12 + 15 = 27$  values are all equal.

The several identities can of course be verified by means of the relations between the nine angles, or rather the derived sine- and cosine-relations

$$C = ab - AB,$$

$$c = aB + bA, \&c.$$



Thus, as regards the two forms of  $S\left(\begin{smallmatrix} a, & b, & c \\ d, & e, & f \end{smallmatrix}\right)$ , the identity to be verified may be written

$$c(d^2Ab + e^2Ba - cFde) = f(a^2De + b^2Ed - fCab).$$

Proceeding to reduce the factor  $a^2De + b^2Ed - fCab$ , if we first write herein  $f = eD + dE$ , it becomes

$$a^2De + b^2Ed - (eD + dE)Cab,$$

which is

$$= aDe(a - bC) + bEd(b - aC),$$

and then writing  $C = ab - AB$ , we have  $a - bC = a(1 - b^2) + bAB = B(aB + bA) = Bc$ ; and, similarly,  $b - aC = Ac$ ; whence the term is  $= c(aeBD + bdAE)$ ; or, in the equation to be verified, the right-hand side is  $= cf(aeBD + bdAE)$ , and by a similar reduction, the left-hand side is found to have the same value.

The paper contains various other interesting results.

## 686.

## ON A FUNCTIONAL EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 315–325; *Proceedings of the London Mathematical Society*, vol. ix. (1878), p. 29.]

I was led by a hydrodynamical problem to consider a certain functional equation viz. writing for shortness  $x_1 = \frac{ax+b}{cx+d}$ , this is

$$\phi x - \phi x_1 = (x - x_1) \frac{Ax + B}{Cx + D}.$$

I find by a direct process, which I will afterwards explain, the solution

$$\phi x = \frac{A}{C}x + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \int_0^\infty \frac{\sin \xi t \sin \eta t dt}{\sin \zeta t \sinh \pi t},$$

where  $\zeta$  is a constant, but  $\xi, \eta$  are complicated logarithmic functions of  $x$  ( $\xi, \eta, \zeta$  depend also on the quantities  $a, b, c, d, C, D$ );  $\sinh \pi t$  denotes as usual the hyperbolic sine,  $\frac{1}{2}(e^{\pi t} - e^{-\pi t})$ .

The values of  $\xi, \eta, \zeta$  are given by the formulæ

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

$$\mathfrak{a} = ax + b, \quad \mathfrak{b} = -dx + b,$$

$$\mathfrak{c} = cx + d, \quad \mathfrak{d} = cx - a,$$

$$W = C\mathfrak{a} + D\mathfrak{c},$$

$$Z = C\mathfrak{b} + D\mathfrak{d},$$

$$R = \lambda\mathfrak{c} + \lambda\mathfrak{d},$$

$$S = -\mathfrak{c} - \mathfrak{d},$$

$$R' = W + \frac{1}{\lambda}Z,$$

$$S' = -W - \lambda Z,$$

which determine  $\lambda, R, S, R', S'$  and then

$$\xi = \frac{1}{2} \log \lambda, \quad \eta = \frac{1}{2} \log \frac{RR'}{SS'}, \quad \zeta = \frac{1}{2} \log \frac{RR'}{SS'}, \quad \zeta = \frac{1}{2} \log \lambda.$$

There is some difficulty as to the denominator factor  $\sin \zeta t$ , which becomes  $= 0$  for the series of values  $t = \frac{n\pi}{\zeta}$ , but this is a point which I do not enter into.

I will in the first instance verify the result. Writing  $x_1$  in place of  $x$ , and taking  $\xi_1, \eta_1$  to denote the corresponding values of  $\xi, \eta$ , it will be shown that

$$\xi_1 = \xi, \quad \eta_1 = \eta + 2\zeta, \quad \text{see post, (1).}$$

Hence in the difference  $\phi x - \phi x_1$  we have the integral

$$\int \frac{\sin \xi t \{ \sin \eta t - \sin(\eta + 2\zeta)t \} dt}{\sin \zeta t \sinh \pi t}$$

(where and in all that follows the limits are  $0, \infty$  as before); here, since

$$\sin \eta t - \sin(\eta + 2\zeta)t = -2 \sin \zeta t \cos(\eta + \zeta)t,$$

the factor  $\sin \zeta t$  divides out, and the numerator is

$$= -2 \sin \xi t \cos(\eta + \zeta)t,$$

which is

$$= \sin(\eta + \zeta - \xi)t - \sin(\eta + \zeta + \xi)t.$$

Hence the integral in question is

$$- \int \frac{\sin(\eta + \zeta - \xi)t dt}{\sinh \pi t} + \int \frac{\sin(\eta + \zeta + \xi)t dt}{\sinh \pi t}.$$

Now we have in general

$$\frac{1}{1 + \exp .a} - \frac{1}{2} = - \int \frac{\sin at dt}{\sinh \pi t},$$

(this is, in fact, Poisson's formula

$$\frac{1}{1 + e^a} - \frac{1}{2} = -2 \int \frac{\sin 2n \log \frac{1}{k} \log k \cdot t dt}{e^{2\pi t} - 1}$$

in the second Memoir on the distribution of Electricity, &c., *Mém. de l'Inst.*, 1811, p. 223); and hence the value is

$$\frac{1}{1 + \exp .(\eta + \zeta - \xi)} - \frac{1}{1 + \exp .(\eta + \zeta + \xi)},$$

or since

$$\eta + \zeta = \frac{1}{2} \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'}, \quad \xi = \frac{1}{2} \log \lambda,$$

we have

$$\eta + \zeta - \xi = \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'} = \log \lambda \frac{R}{S},$$

$$\eta + \zeta + \xi = \log \lambda + \frac{1}{2} \log \frac{R'^2}{S'^2} = \log \lambda \frac{R'}{S'},$$

and the value is thus

$$\frac{1}{1 + \lambda \frac{R}{S}} - \frac{1}{1 + \lambda \frac{R'}{S'}} = \frac{(RS' - R'S)\lambda}{(\lambda R + S)(\lambda R' + S')}.$$

Hence, from the assumed value of  $\phi x$ , we obtain

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \frac{(RS' - R'S)\lambda}{(\lambda R + S)(\lambda R' + S')}.$$

We have

$$RS' - R'S = \frac{(\lambda - 1)(\lambda^2 - 1)}{\lambda}(dC - cD)\{cx^2 + (d - a)x - b\},$$

$$R\lambda + S = (\lambda^2 - 1)(cx + d), \quad \text{see post, (2),}$$

or since

$$R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D),$$

this is

$$\phi x - \phi x_1 = (x - x_1) \frac{A}{C} + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \frac{cx^2 + (d - a)x - b}{(a + d)\lambda} \lambda'.$$

But from the value of  $\lambda$ ,

$$\frac{\lambda}{\lambda^2 - 1} = \frac{ad - bc}{(a + d)\sqrt{(a - d)^2 + 4bc}},$$

and the equation thus is

$$\phi x - \phi x_1 = (x - x_1) \left\{ \frac{A}{C} - \frac{AD - BC}{C(Cx + D)} \right\} = (x - x_1) \frac{Ax + B}{Cx + D},$$

as it should be.

(1) For the foregoing values of  $\xi_1, \eta_1$ , we require  $R_1, S_1, R'_1, S'_1$ , the values which  $R, S, R', S'$  assume on writing therein  $x_1$  for  $x$ . We have

$$R_1 = \lambda(cx_1 + d) + (cx_1 - a),$$

$$S_1 = -(cx_1 + d) - \lambda(cx_1 - a);$$

substituting for  $x_1$  its value, we find

$$R_1(cx + d) = (a + d)\lambda(cx + d) - (ad - bc)(\lambda + 1),$$

or writing herein

$$ad - bc = \frac{(a + d)^2 x}{\lambda + 1},$$

this is

$$R_1(cx + d) = \frac{a + d}{\lambda + 1} R,$$

and similarly

$$S_1(cx + d) = \frac{a + d}{\lambda + 1} S.$$

We have in like manner

$$R'_1 = W_1 + \frac{1}{\lambda}Z_1, \text{ where } W_1 = C(ax_1 + b) + D(cx_1 + d),$$

$$S'_1 = -W_1 - \lambda Z_1, \text{ where } Z_1 = C(-dx_1 + b) + D(cx_1 - a).$$

Substituting for  $x_1$  its value, we find

$$W_1(cx + d) = C[(a + d)(ax + b) - (ad - bc)x] + D[(a + d)(cx + d) - (ad - bc)],$$

$$Z_1(cx + d) = C[-(ad - bc)x] + D[-(ad - bc)],$$

hence, substituting for  $ad - bc$  as before,

$$W_1(cx + d) = \frac{a + d}{(\lambda + 1)^2} \{(\lambda + 1)^2 W - (a + d)\lambda(Cx + D)\},$$

$$Z_1(cx + d) = \frac{a + d}{(\lambda + 1)^2} \{-(a + d)\lambda(Cx + D)\},$$

whence without difficulty

$$\frac{R'_1}{R_1} = \frac{R'}{R}; \quad \frac{S'_1}{S_1} = \frac{S'}{S} \cdot \frac{1}{\lambda},$$

consequently

$$\frac{R_1 R'_1}{S_1 S'_1} = \frac{RR'}{SS'} \cdot \lambda, \quad \eta_1 = \log \lambda + \eta = \eta + 2\zeta, \quad \xi_1 = \xi,$$

which are the formulæ in question.

(2) For the value of  $RS' - R'S$ , we have

$$\begin{aligned} RS' - R'S &= (\lambda c + d)(-W - \lambda Z) - (-\lambda d - c)(W + \frac{1}{\lambda}Z) \\ &= (-\lambda^2 + 1)cZ + (\lambda + 1)\{(d - c)W - dR\} \div \frac{1}{\lambda} \\ &= -(\lambda - 1) \left\{ \left( 1 + \lambda + \frac{1}{\lambda} \right) cZ + (c - d)W + dZ \right\}; \end{aligned}$$

or substituting for  $\lambda + \frac{1}{\lambda}$ ,  $W$  and  $Z$  their values, this is

$$\begin{aligned} &= \frac{-(\lambda - 1)}{ad - bc} \{(a^2 + d^2 + ad + bc)c(bC + dD) \\ &\quad + (ad - bc)[(c - d)(aC + cD) + d(bC + dD)]\}. \end{aligned}$$

In the term in  $\{ \}$ , the coefficient of  $C$  is

$$\begin{aligned} &[(a^2 + d^2 + ad + bc)b + (ad - bc)a]c - d(a - b)(ad - bc) \\ &= (a + d)(db - bd)c - (a + d)dx(ad - bc), \end{aligned}$$

and similarly the coefficient of  $D$  is

$$\begin{aligned} & [(a^2 + d^2 + ad + bc)d + (ad - bc)c]c - d(c - d)(ad - bc) \\ &= (a + d)(ad - bc)c - (a + d)d(ad - bc). \end{aligned}$$

Hence the whole term in  $\{ \}$  is

$$= (a + d)\{[(db - bd)c - d(ad - bc)x]C + [(ad - cb)c - d(ad - bc)]D\},$$

which is readily reduced to

$$(a + d)(ad - bc)(-dC + cD);$$

also

$$ad - bc = (a + d)\{cx^2 + (d - a)x - b\};$$

so that we have

$$RS' - R'S = \frac{(\lambda - 1)(\lambda^2 - 1)}{\lambda}(dC - cD)\{cx^2 + (d - a)x - b\},$$

which is the required value of  $RS' - R'S$ ; and there is no difficulty in obtaining the other two formulæ,

$$R\lambda + S = (\lambda^2 - 1)(cx + d),$$

$$R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D);$$

the verification is thus completed.

To show how the formula was directly obtained, we have

$$\begin{aligned} \frac{Ax + B}{Cx + D} &= \frac{A}{C} - \frac{AD - BC}{C} \cdot \frac{1}{Cx + D} \\ &= \frac{A}{C} + \beta x, \text{ suppose;} \end{aligned}$$

the equation then is

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) + (x - x_1)\beta x.$$

Hence, if  $x_1, x_2, x_3, \dots$  denote the successive functions  $\psi x, \psi^2 x, \psi^3 x$ , &c., we have

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) + (x - x_1)\beta x,$$

$$\phi x_1 - \phi x_2 = \frac{A}{C}(x_1 - x_2) + (x_1 - x_2)\beta x_1,$$

whence adding, and neglecting  $\phi x_\infty$  and  $x_\infty$ , we have

$$\phi x = \frac{A}{C}x + [(x - x_1)\beta x + (x_1 - x_2)\beta x_1 + (x_2 - x_3)\beta x_2 + \dots],$$

where the term in  $[ ]$ , regarding therein  $x_1, x_2, x_3, \dots$  as given functions of  $x$ , is itself a given function of  $x$ ; and it only remains to sum the series.

Starting from

$$x_1 = \psi x = \frac{ax + b}{cx + d},$$

and writing

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

then the  $n$ th function is given by the formula

$$\begin{aligned} x_n = \psi^n x &= \frac{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)}{(\lambda^{n+1} - 1)(cx + d) + (\lambda^n - \lambda)(cx - a)} \\ &= \frac{(\lambda^{n+1} - 1)a + (\lambda^n - \lambda)b}{(\lambda^{n+1} - 1)c + (\lambda^n - \lambda)d} \quad (\text{numerator}) \div (\text{denominator}) \\ &= \frac{\lambda^n P + Q}{\lambda^n R + S}, \end{aligned}$$

if  $P = \lambda a + b$ ,  $Q = -a - \lambda b$ , and  $R = \lambda c + d$ ,  $S = -c - \lambda d$ , as before.

I stop to remark that  $\lambda$  being real, then if  $\lambda > 1$  we have  $\lambda^n$  very large for  $n$  very large, and  $x_n = \frac{P}{R}$  which is independent of  $n$ ; the value in question is

$$x_n = \frac{\lambda(ax + b) + (-dx + b)}{\lambda(cx + d) + (cx - a)},$$

which, observing that the equation in  $\lambda$  may be written

$$\frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a},$$

is, in fact, independent of  $x$ , and  $= \frac{\lambda a - d}{c(\lambda + 1)}$  or  $\frac{b(\lambda + 1)}{\lambda d - a}$ : we have  $x_{n-1} = x_n$ , or calling each of these two equal values  $x$ , we have

$$x = \frac{ax + b}{cx + d},$$

which is the same equation as is obtainable by the elimination of  $\lambda$  from the equations

$$x = \frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a}.$$

The same result is obtained by taking  $\lambda < 1$  and consequently  $x_n = \frac{Q}{S}$ .

We find

$$x_{n-1} - x_n = \frac{\lambda^{n-1}P + Q}{\lambda^{n-1}R + S} - \frac{\lambda^n P + Q}{\lambda^n R + S},$$

where

$$PS - QR = -(\lambda^2 - 1)(ad - bc) = -(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\},$$

and therefore

$$x_{n-1} - x_n = \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(\lambda^{n-1}R + S)(\lambda^n R + S)}.$$

Also

$$\beta x_n = -\frac{AD - BC}{C} \cdot \frac{1}{Cx_n + D},$$

where

$$Cx_n + D = \frac{C(\lambda^n P + Q) + D(\lambda^n R + S)}{\lambda^n R + S} = \frac{R'\lambda^n + S'}{\lambda^n R + S},$$

where

$$\begin{aligned} R' &= CP + DR, = C\left(-b + \frac{1}{\lambda}\right) + D\left(c + \frac{1}{\lambda}\right), \\ S' &= CQ + DS, = C(-b - \lambda) + D(-c - d\lambda); \end{aligned}$$

viz.

$$R' = W + \frac{1}{\lambda}Z, \quad S' = -W - \lambda Z,$$

where  $Z$  and  $W$  denote  $aC + cD$  and  $bC + dD$  as before.

We hence obtain

$$\begin{aligned} (x_{n-1} - x_n)\beta x_n &= \frac{-(AD - BC)}{C} \cdot \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(R\lambda^n + S)(R'\lambda^n + S')} \\ &= \frac{-(AD - BC)}{C} \cdot \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(dC - cD)(R\lambda^n + S)(R'\lambda^n + S')}, \end{aligned}$$

or, substituting for  $RS' - R'S$  its value in the denominator, this is

$$\begin{aligned} (x_{n-1} - x_n)\beta x_n &= \frac{AD - BC}{C} \cdot \frac{-(ad - bc)(\lambda^n - 1)(RS' - R'S)\lambda^n}{(a + d)\lambda(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')} \\ &= \frac{\sqrt{(a - d)^2 + 4bc}(AD - BC)(RS' - R'S)\lambda^n}{C(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')}, \end{aligned}$$

and thence

$$\phi x = \frac{A}{C}x + \Sigma \frac{\sqrt{(a - d)^2 + 4bc}(AD - BC)(RS' - R'S)\lambda^n}{C(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')},$$

the summation extending from 1 to  $\infty$ .

Now the before-mentioned integral formula gives

$$\begin{aligned} \frac{1}{1 + k\lambda^n} - \frac{1}{2} &= - \int \frac{\sin(n \log \lambda + \log k)t \, dt}{\sinh \pi t}, \\ \frac{1}{1 + k'\lambda^n} - \frac{1}{2} &= - \int \frac{\sin(n \log \lambda + \log k')t \, dt}{\sinh \pi t}. \end{aligned}$$

Taking the difference, and then writing  $k = \frac{R}{S}$ ,  $k' = \frac{R'}{S'}$ , we have under the integral sign

$$\sin \left( n \log \lambda + \log \frac{R}{S} \right) t - \sin \left( n \log \lambda + \log \frac{R'}{S'} \right) t,$$



which is

$$= 2 \sin \frac{1}{2} \left( \log \frac{RR'}{SS'} \right) t \cos \left( n \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'} \right) t,$$

which attending to the before-mentioned values of  $\xi$ ,  $\eta$ ,  $\zeta$  is

$$= 2 \sin \zeta t \cos(2n\zeta - \xi + \eta)t,$$

and the formula thus is

$$\frac{S}{R\lambda^n + S} - \frac{S'}{R'\lambda^n + S'} = \frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = -2 \int \frac{\sin \zeta t \cos(2n\zeta - \xi + \eta)t dt}{\sinh \pi t}.$$

We have here

$$\cos(2n\zeta - \xi + \eta)t = \cos 2n\zeta t \cos(\eta - \xi)t - \sin 2n\zeta t \sin(\eta - \xi)t,$$

whence summing from 1 to  $\infty$  by means of the formulæ

$$\cos 2\zeta t + \cos 4\zeta t + \dots = -\frac{1}{2},$$

$$\sin 2\zeta t + \sin 4\zeta t + \dots = \frac{1}{2} \cot \zeta t,$$

(which series however are not convergent), the numerator under the integral sign becomes

$$- \sin \zeta t \{ -\cos(\eta - \xi)t - \cot \zeta t \sin(\eta - \xi)t \},$$

which is

$$= \frac{\sin \xi t \sin \eta t}{\sin \zeta t},$$

and the formula thus is

$$\Sigma \frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = - \int \frac{\sin \xi t \sin \eta t}{\sin \zeta t \sinh \pi t} dt,$$

and we therefore find

$$\phi x = \frac{A}{C}x + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(cD - dC)} \int \frac{\sin \xi t \sin \eta t dt}{\sin \zeta t \sinh \pi t},$$

which is the result in question.

The solution is a particular one; calling it for a moment  $(\phi x)$ , then, if the general solution be  $\Phi x = \phi x + \varpi x$ , it at once appears that we must have  $\varpi x = \varpi x_1$ ; and as it has been shown that  $\frac{RR'}{SS'}$  is a function of  $x$  which remains unaltered by the change of  $x$  into  $x_1$ , this is satisfied by assuming  $\varpi x = f\left(\frac{RR'}{SS'}\right)$ , an arbitrary function of  $\frac{RR'}{SS'}$ . Hence we may to the foregoing expression of  $\phi x$  add this term  $f\left(\frac{RR'}{SS'}\right)$ .

*Postscript.* The new formula

$$x_n = \frac{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)}{(\lambda^{n+1} - 1)(cx + d) + (\lambda^n - \lambda)(cx - a)},$$

where

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

for the  $n$ th repetition of  $\psi x$ ,  $= \frac{ax+b}{cx+d}$ , is a very interesting one. It is to be remembered that, when  $n$  is even the numerator and denominator each divide by  $\lambda-1$ , but when  $n$  is odd they each divide by  $\lambda^2-1$ ; after such division, then further dividing by a power of  $\lambda$ , they each consist of terms of the form  $\lambda^k + \frac{1}{\lambda^k}$ , that is, they are each of them a rational function of  $\lambda + \frac{1}{\lambda}$ . Substituting and multiplying by the proper power of  $ad-bc$ , the numerator and denominator become each of them a rational and integral function of  $a, b, c, d$  of the order  $n+1$  when  $n$  is even, but of the order  $n$  when  $n$  is odd; in the former case, however, the numerator and denominator each divide by  $a+d$ , so that ultimately, whether  $n$  be even or odd, the order is  $n$  as it should be.

For example, when  $n=2$ , the value is

$$\frac{(\lambda^3-1)a + (\lambda^2-\lambda)b}{(\lambda^3-1)c + (\lambda^2-\lambda)d} = \frac{(\lambda^2+\lambda+1)a + \lambda b}{(\lambda^2+\lambda+1)c + \lambda d} = \frac{(\lambda + \frac{1}{\lambda} + 1)a + b}{(\lambda + \frac{1}{\lambda} + 1)c + d},$$

or, as this may be written,

$$= \frac{(\lambda + \frac{1}{\lambda} + 2)a - a + b}{(\lambda + \frac{1}{\lambda} + 2)c - c + d},$$

where, observing that

$$\lambda + \frac{1}{\lambda} + 2 = \frac{(a+d)^2}{ad-bc}, \quad -a+b = -(a+d) \cdot \frac{ad-bc}{(a+d)^2},$$

the numerator and denominator each divide by  $a+d$ , and the final value is

$$= \frac{(a+d)(ax+b) - (ad-bc)x}{(a+d)(cx+d) - (ad-bc)} = \frac{(a^2+bc)x + b(a+d)}{c(a+d)x + bc + d^2},$$

which is the proper value of  $\psi^2 x$ . But, when  $n=3$ , the value is

$$\frac{(\lambda^4-1)a + (\lambda^3-\lambda)b}{(\lambda^4-1)c + (\lambda^3-\lambda)d} = \frac{(\lambda^2+1)a + \lambda b}{(\lambda^2+1)c + \lambda d}$$

and this is

$$= \frac{(a^2+d^2+2bc)(ax+b) + (ad-bc)(-dx+b)}{(a^2+d^2+2bc)(cx+d) + (ad-bc)(cx-a)}$$

or finally

$$= \frac{(a^3+2abc+bcd)x + b(a^2+ad+bc+d^2)}{c(a^2+ad+bc+d^2)x + (abc+2bcd+d^3)},$$

which is the proper value of  $\psi^3 x$ .

## 687.

NOTE ON THE FUNCTION  $\psi x = a^2(c - x) \div \{c(c - x) - b^2\}$ .[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 338–340.]

Starting from the general form

$$\psi x = \frac{\alpha x + \beta}{\gamma x + \delta},$$

we have

$$\psi^n x = \frac{(\lambda^{n+1} - 1)(\alpha x + \beta) + (\lambda^n - \lambda)(-\delta x + \beta)}{(\lambda^{n+1} - 1)(\gamma x + \delta) + (\lambda^n - \lambda)(\gamma x - \alpha)},$$

where

$$\lambda + \frac{1}{\lambda} = \frac{\alpha^2 + \delta^2 + 2\beta\gamma}{\alpha\delta - \beta\gamma}.$$

For the function in question

$$\psi x = \frac{a^2(c - x)}{c(c - x) - b^2},$$

(a form which presents itself in the problem of the distribution of electricity upon two spheres), the values of  $\alpha, \beta, \gamma, \delta$  are

$$\alpha = -a^2, \quad \beta = a^2c, \quad \gamma = -c, \quad \delta = c^2 - b^2;$$

the equation for  $\lambda$  therefore is

$$\lambda + \frac{1}{\lambda} = \frac{a^4 + (c^2 - b^2)^2 - 2a^2c^2}{a^2b^2},$$

or, what is the same thing,

$$\frac{(\lambda + 1)^2}{\lambda} = \frac{(a^2 + b^2 - c^2)^2}{a^2b^2}.$$

Suppose that  $a, b, c$  are the sides of a triangle the angles whereof are  $A, B, C$ ; then  $c^2 = a^2 + b^2 - 2ab \cos C$ , or we have

$$\frac{(\lambda + 1)^2}{\lambda} = 4 \cos^2 C;$$

or, writing this under the form

$$\sqrt{\lambda} + \frac{1}{\sqrt{\lambda}} = 2 \cos C,$$

the value of  $\lambda$  is at once seen to be  $= e^{2iC}$ ; and it is interesting to obtain the expression of the  $n$ th function in terms of the sides and angles of the triangle.

The numerator and the denominator are

$$\lambda^n P + Q,$$

$$\lambda^n R + S,$$

where

$$P = \lambda(\alpha x + \beta) + (-\delta x + \beta), \quad R = \lambda(\gamma x + \delta) + \gamma x - \alpha,$$

$$Q = -(\alpha x + \beta) - \lambda(-\delta x + \beta), \quad S = -(\gamma x + \delta) - \lambda(\gamma x - \alpha).$$

Hence, writing the numerator and the denominator in the forms

$$\lambda^{\frac{1}{2}n} P + \lambda^{-\frac{1}{2}n} Q,$$

these are

$$(P + Q) \cos nC + (P - Q)i \sin nC,$$

viz. they are

$$(R + S) \cos nC + (R - S)i \sin nC;$$

$$(\lambda - 1)(\alpha + \delta)x \cos nC + (\lambda + 1)\{(\alpha - \delta)x + 2\beta\}i \sin nC,$$

$$-(\lambda - 1)(\alpha + \delta) \cdot \cos nC + (\lambda + 1)\{2\gamma x - (\alpha - \delta)\}i \sin nC,$$

or, observing that  $\frac{\lambda - 1}{\lambda + 1} = i \tan C$  and removing the common factor  $i(\lambda + 1)$ , they may be written

$$\tan C(\alpha + \delta)x \cos nC + \{(\alpha - \delta)x + 2\beta\} \sin nC,$$

$$\tan C(\alpha + \delta) \cdot \cos nC + \{2\gamma x - (\alpha - \delta)\} \sin nC.$$

Substituting for  $\alpha, \beta, \gamma, \delta$  their values, these are

$$\tan C\{(c^2 - a^2 - b^2)x \cos nC\} + \{(b^2 - a^2 - c^2)x + 2a^2c\} \sin nC,$$

$$-a - \tan C\{(c^2 - a^2 - b^2) \cdot \cos nC\} + \{-2cx - (b^2 - a^2 - c^2)\} \sin nC,$$

$$= \tan C\{-ab \cos C \cdot x \cos nC\} + \{-ac \cos B \cdot x + a^2c\} \sin nC,$$

$$\tan C\{-ab \cos C \cdot x \cos nC\} + \{-cx + ac \cos B\} \sin nC,$$

$$= x\{-ab \sin C \cos nC - ac \cos B \sin nC\} + a^2c \sin nC,$$

$$-cx \sin nC + \{ac \cos B \sin nC - ab \sin C \cos nC\};$$

or, writing herein  $b \sin C = c \sin B$ , these are

$$\begin{aligned} & -acx\{\sin B \cos nC + \cos B \sin nC\} + a^2c \sin nC, \\ & -cx \sin nC + ac\{\cos B \sin nC - \sin B \cos nC\}, \end{aligned}$$

whence finally

$$\psi^n x = \frac{a^2 \sin nC - ax \sin(nC + B)}{a \sin(nC - B) - x \sin nC}.$$

As a verification, writing  $n = 1$ , we have

$$\psi x = \frac{a^2 \sin C - ax \sin A}{a \sin(C - B) - x \sin C} = \frac{a^2 c - acx \frac{\sin A}{\sin C}}{a \frac{\sin(C - B)}{\sin C} \cdot c - cx},$$

or observing that

$$ac \frac{\sin(C - B)}{\sin C} = c^2 - b^2,$$

(for this is  $\sin A = \sin(C - B) \sin^2 C - \sin^2 B$ ), we have

$$\psi x = \frac{a^2(c - x)}{c^2 - b^2 - cx},$$

as it should be. If in the formula for  $\psi^n x$  we write  $x = 0$ , we have a formula given in the Senate-House Problems, January 14, 1878: it was thus that I was led to investigate the general expression.

## 688.

## GEOMETRICAL CONSIDERATIONS ON A SOLAR ECLIPSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 340–347.]

I consider, from a geometrical point of view, the phenomena of a solar eclipse over the earth generally; attending at present only to the penumbral cone, the vertex of which I denote by  $V$ . It is convenient to regard the earth as fixed, and the sun and moon as moving each of them with its proper motion, and also with the diurnal motion. The penumbral cone meets the earth's surface in a curve which may be called the penumbral curve; viz. when the cone is not completely traversed by the earth's surface, (that is, when only some of the generating lines of the cone meet the earth's surface), the penumbral curve is a single (convex or hour-glass-shaped) oval; separated, as afterwards mentioned, into two parts, one of them lying away from the sun, and having no astronomical significance; but when the cone is completely traversed by the earth's surface, then the penumbral curve consists of two separate (convex) ovals; one of them lying away from the sun and having no astronomical significance, the other lying towards the sun. The intermediate case is when the cone just traverses the earth's surface, or is touched internally by the earth's surface; the penumbral curve is then a figure of eight, one portion of which lies away from the sun, and has no astronomical significance: there is another limiting case when the cone is touched externally by the earth's surface, the penumbral curve being then a mere point.

It is necessary to consider on the earth's surface a curve which may for shortness be termed the horizon; viz. this is the curve of contact of the cone, vertex  $V$ , circumscribed about the earth; it is a small circle nearly coincident with the great circle, which is the intersection by a plane through the centre of the earth at right angles to the line from this point to the centre of the sun.

Regarding  $V$  as a point in the heavens, capable of being viewed notwithstanding the interposition of the moon; the horizon, as above defined, is the curve separating

the portions of the earth's surface for which  $V$  is visible and invisible respectively. The horizon does or does not meet the penumbral curve, according as this last consists of a single oval or of two distinct ovals; viz. in the latter case the horizon lies between the two ovals, in the former case the horizon traverses the area of the oval (separating this area into two parts), thus meeting the oval, or penumbral curve, in two points, or say these points separate the oval into two parts; from any point of the one part  $V$  is visible, from any point of the other part  $V$  is invisible; and from each of the two points themselves  $V$  is visible as a point on the horizon in the ordinary sense of the word; that is, there is an exterior contact of the sun and moon visible on the horizon. It is to be observed that, in the limiting cases where the penumbral curve is a mere point and a figure of eight respectively, the horizon passes through the mere point and through the node of the figure of eight respectively.

The two points of intersection of the penumbral curve with the horizon may for shortness be termed *critic points*. The lines which present themselves in a diagram of a solar eclipse, (see *Nautical Almanac*;) are the “northern and south lines of simple contact,” say for shortness the “limits”; viz. these are the envelope or, geometrically, a portion of the envelope of the penumbral curve; and the lines of “eclipse begins or ends at sunrise or sunset,” say for shortness the *critic lines*; viz. these are the locus of the *critic points*.

The point  $V$  considered as a point in the heavens is a point occupying a position intermediate between those of the centres of the sun and moon; hence referring it to the surface of the earth by means of a line drawn from the centre, its position on the earth's surface is nearly coincident with that point to which the sun is then vertical; and its motion on the earth's surface is from east to west approximately along the parallel of latitude = sun's declination, and with a velocity of approximately  $15^\circ$  per hour. For any given position of  $V$  on the earth's surface, describing with a given angular radius nearly  $= 90^\circ$  a small circle (nearly a great circle), this is the horizon; as  $V$  moves upon the surface of the earth, the horizon envelopes a curve which is very nearly a parallel, angular radius = sun's declination (there are two such curves in the northern and southern hemispheres respectively, but I attend only to one of them in the proper hemisphere, as will be explained), say this is the horizon-envelope; the horizon in each of its successive positions is thus a curvilinear tangent (nearly a great circle) to this horizon-envelope. If for a given position of  $V$ , and also for the consecutive position we consider the corresponding horizons, these intersect in a point  $K$  on the horizon-envelope, and the horizon for  $V$  is the circle centre  $V$  and angular radius  $VK$ ;  $K$  is a point which is very nearly upon, and which may be taken to be upon, the meridian through  $V$ ; the horizon may be regarded as a tangent which sweeps round the horizon-envelope; to each position thereof there corresponds a position of  $V$ , and consequently also a penumbral curve; and (when this is a single oval) the horizon meets it in two points, which are the *critic points*. It is to be added that, if for a given position of the horizon we consider as well  $K$  as the opposite point  $K_1$ , (viz.,  $K_1$  lies on the great circle  $KV$ ), then the points  $K$  and  $K_1$  divide the horizon into two portions; for any point on one of these portions

$V$  (considered as a point in the heavens) is rising, for a point on the other of them it is setting; and for the points  $K$  and  $K_1$  respectively it is moving horizontally; that is, first rising and then setting, or vice versa.

A solar eclipse is of one of two classes; viz. either the penumbral cone completely traverses the earth, so that towards the middle of the eclipse the penumbral curve consists of two separate ovals: or the penumbral cone does not completely traverse the earth, so that throughout the eclipse the penumbral curve consists of a single oval only. In the former case, we have to consider the commencement, during which the penumbral curve passes from a mere point to a figure of eight: the middle, during which it passes from a figure of eight through two ovals to a figure of eight: and the termination, during which it passes from a figure of eight to a mere point. In the latter case, we consider the whole eclipse during which the penumbral curve passes from a mere point through a single oval to a mere point.

In an eclipse of the first class: for the commencement, the penumbral curve is at first a mere point (point of first contact); it then becomes a convex oval, each oval in the first instance inclosing the preceding ones, so that there is not any intersection of two consecutive ovals. We come at last to an oval which is touched north by its consecutive oval, and to an oval which is touched south by its consecutive oval (I presume that the contacts north and south do not take place on the same oval, but I am not sure); and after this, the ovals assume the hour-glass form, each oval intersecting the consecutive oval in two points north and two points south; the ovals thus beginning to form an envelope or limit. There are on each of the ovals two critic points, and we have thus a critic curve commencing at the mere point (point of first contact) and extending in each direction from this point. The point, where an oval is touched by the consecutive oval, is not so far as appears a critic point; that is, the critic curve does not at this point unite itself with the envelope or limit. But the critic curve comes subsequently to unite itself each way with the limit; and, since clearly it cannot intersect the limit, it will at each of these points touch the limit; that is, we have a critic curve extending each way from the point of first contact until it touches the northern limit and until it touches the southern limit. Observe that the penumbral curve, as being at first a mere point or an indefinitely small oval, does not at first contain within itself the point  $K$  or  $K_1$ ; it can only come to do this by passing through a position where the curve passes through  $K$  or  $K_1$ ; viz.  $K$  or  $K_1$  would then be a critic point; and I assume for the present that this does not take place. The critic curve at the point of first contact is a curve "eclipse begins at sunrise," and as not coming to pass through a point  $K$  or  $K_1$ , it cannot alter its character; that is, the critic curve, as extending each way from the point of first contact until it comes to touch the northern and southern limits respectively, is a curve "eclipse begins at sunrise"; at the terminal points in question, there is a mere contact of the sun and moon, so that they are points, where the eclipse begins and simultaneously ends at sunrise. Continuing the series of ovals until we arrive at the figure of eight, there are on each of them two critic points, which ultimately unite in the node of the figure of eight; these constitute a critic curve, extending each way from the node of the figure of eight to



the contacts with the northern and southern limits respectively. There is, as before, no passage through a point  $K$  or  $K_1$ , the curve in question thus retains throughout the same character; and by consideration of the two terminal points it at once appears that it is a curve “eclipse ends at sunrise.” The above-mentioned critic curves form together an oval touching the northern and southern limits respectively; say this is the sunrise oval.

The termination of the eclipse is similar to this, only the events happen in the reverse order; we have a critic line starting from the node of the figure of eight and extending each way until it comes to touch the northern and southern limits respectively, viz. this is the line “eclipse begins at sunset”; and then, extending each way from the points of contact to reunite itself at the point of last contact, this being the line “eclipse ends at sunset,” and the two portions together form an oval touching the northern and southern limits respectively; say this is the sunset oval. It is to be noticed that certain portions of the two limits are generated as the envelope of the penumbral curve during the commencement and during the termination of the eclipse.

For the middle of the eclipse; the penumbral curve, in the first instance a figure of eight, breaks up into two ovals, but only one of these is attended to; and ultimately the oval unites itself with another oval so as to give rise to a new figure of eight. There is thus throughout the middle of the eclipse a single oval: this has, north and south, an envelope which joins itself on to the portions enveloped during the latter part of the commencement and the former part of the termination of the eclipse, and constitutes therewith the northern and southern limits respectively, viz. each of these is considered as extending from a point of contact with the sunrise oval to a point of contact with the sunset oval.

The line  $K_1VK$ , or say the meridian line through  $V$ , travels westwardly, while the penumbral curve travels eastwardly; the two come to touch each other, and there are then two intersections which ultimately come to the northern and southern limits respectively: the locus of these is a line of “eclipse commences at midday”; as the motion continues, the points of intersection move away from the two limits respectively and ultimately unite at the point where the line  $KVK_1$  again touches the penumbral curve; the locus is the line of “eclipse terminates at midday,” the two lines together forming an oval which touches the northern and southern limits respectively and which may be termed the midday oval. In all that precedes, no distinction has been made between the two portions of the horizon-envelope, or the points  $K$  and  $K_1$ , and either curve and point indifferently may be alone attended to.

Considering now an eclipse of the second kind, the penumbral curve is at first a mere point (the point of first contact) and it then becomes an oval, the successive ovals not at first intersecting each other, but each oval inclosing within itself the preceding ones. Any oval is met by the corresponding horizon in two points  $P$  and  $P'$ , at first coinciding with each other at the point of first contact, and then separating from each other, one of them, say  $P$ , moving down towards and ultimately arriving at one of the horizon-envelopes, say to fix the ideas the southern one (which curve

is henceforth selected as being, and is called, the horizon-envelope, and the points on this curve are taken to be the points  $K$ ), viz.  $P$  is then a point on the penumbral curve, I call it  $K_1$ . The successive ovals will in the meantime have begun to intersect each other so as to give rise to a northern limit; this will touch the critic line (locus of  $P, P'$ ), and we have a portion of the critic line extending from the point of first contact, in one direction to the point of contact with the northern limit, and in the other direction to the point  $K_1$  on the horizon-envelope: this is the line “eclipse begins at sunrise.” As the horizon continues to sweep on, the other point  $P'$ , which has not yet reached the horizon-envelope, will gradually approach and ultimately arrive at the horizon-envelope, say at the point  $K_2$ ; we have thus a second portion of the critic line extending from the contact with the northern envelope to the point  $K_2$ ; this is the line “eclipse ends at sunrise.” The horizon continuing to sweep on, the point  $P$  beginning with the position  $K_1$ , which is now on the other side of the point of contact of the horizon with the horizon-envelope, will trace out a portion of the critic curve extending from  $K_1$  to a second point of contact with the northern limit; this will be the line of “eclipse begins at sunset.” And, finally, the point  $P$  from the last-mentioned point of contact, and the point  $P'$  from its position  $K_2$ , which is now on the other side of the point of contact of the horizon with the horizon-envelope, (that is,  $P, P'$  have now each passed through the point of contact of the horizon with the horizon-envelope, and are both of them on the same side thereof, viz. the side opposite to their original side), will come to unite at the point of the last contact; we have thus a fourth portion of the critic curve extending from  $K_2$  to the second point of contact with the northern limit, viz. this is the line “eclipse ends at sunset.” The description will be more intelligible by means of the figure, in which  $1, 1', 2, 2', \dots, 8, 8'$  represent successive corresponding positions of the points  $P, P'$ , the successive positions of the horizon being given by the right lines  $11', 22', \&c.$ , all of them tangents to the dotted circle or horizon-envelope.

The entire critic line is thus a figure of eight, twice touching the horizon-envelope

and also twice touching the limit. If we consider, as before, the intersections of  $KV$  with the corresponding penumbral curve, this will be a curve extending from  $K_1$  so as to touch the limit, and thence onward to  $K_2$ , the portion from  $K_1$  to the contact with the limit being the line “eclipse begins at transit,” and the portion from the limit to  $K_2$  the line “eclipse ends at transit.” I say “transit” instead of midday, since for a circumpolar place the phenomenon may happen at one or the other transit of the sun over the meridian. It is to be remarked, that the node of the figure of eight is a point, such that the eclipse there begins at sunrise and ends at sunset; this point does not appear to be an important one in the geometrical theory.

The two loops of the critic line may be of very unequal magnitudes, and in particular one of them may actually vanish; viz. the points  $K_1$  and  $K_2$  then coincide together, and the critic curve is a closed cuspidal curve touching the horizon-envelope at the cusp; moreover, instead of two contacts with the limit there is one proper contact, and an improper contact at the cusp, that is, the limit simply passes through the cusp. And through this special separating case, we pass to the case where, instead of the figure of eight, we have a single oval, not touching the horizon-envelope (viz. the points  $K_1, K_2$  have become imaginary), but still touching the limit twice: this is a distinct type for an eclipse of the second class.

And, similarly, in an eclipse of the first class, where the points  $K_1, K_2$  do not in general exist (viz. geometrically they are imaginary), these points may present themselves in the first instance as two coincident points, viz. instead of the sunrise oval or the sunset oval (as the case may be), we have then a cuspidal curve; or they may be two real points, viz. instead of the same oval, we have then a figure of eight touching the horizon-envelope twice, and also touching each of the two limits. These are thus the several cases.

When the Earth traverses the penumbral cone, the critic curve is

1. A pair of ovals;
2. An oval and a cuspidate oval;
3. An oval and a figure of eight.

And when the Earth does not traverse the penumbral cone, the critic curve is

4. A figure of eight;
5. A cuspidate oval;
6. An oval.

To which may be added the transition case which separates 1 and 4, viz. here the Earth just has an internal contact with the penumbral cone, and the critic curve is

7. Two ovals touching each other.

But of course 2, 5, and 7 are so special that they may be disregarded altogether; and 3 and 6 are of rare occurrence. I have not sufficiently examined the conditions for the occurrence of these forms 3 and 6; my attention was called to them, and indeed to the whole theory, by a question proposed by Prof. Adams in the Cambridge Smith's Prize Examination for 1869.

## 689.

ON THE GEOMETRICAL REPRESENTATION OF IMAGINARY  
VARIABLES BY A REAL CORRESPONDENCE OF TWO PLANES.

[From the *Proceedings of the London Mathematical Society*, vol. ix. (1878), pp. 31–39. Read December 13, 1877.]

In my recently published paper, “Geometrical Illustration of a Theorem relating to an Irrational Function of an Imaginary Variable,” *Proceedings of the London Mathematical Society*, t. viii. (1877), pp. 212–214, [627], I remark as follows:—“If we have  $v$  a function of  $u$  determined by an equation  $f(u, v) = 0$ , then to any given imaginary value  $x + iy$  of  $u$  there belong two or more values, in general imaginary, of  $v$ ; and for the complete understanding of the relation between the two imaginary variables we require to know the series of values  $x' + iy'$  which correspond to a given series of values  $x + iy$  of  $v, u$  respectively. We must, for this purpose, take  $x, y$  as the coordinates of a point  $P$  in a plane  $\Pi$ , and  $x', y'$  as the coordinates of a corresponding point  $P'$  in another plane  $\Pi'$ ; and I then proceed to consider the particular case where the equation between  $u, v$  is  $u^2 + v^2 = a^2$ , that is, where

$$(x + iy)^2 + (x' + iy')^2 = a^2.$$

The general case is that of an equation  $(*)(u, 1)^m(v, 1)^n = 0$ , where to each given value, real or imaginary, of  $u$ , there correspond  $n$  real or imaginary values of  $v$ ; and to each given value, real or imaginary, of  $v$ , there correspond  $m$  real or imaginary values of  $u$ . And then, writing  $u = x + iy$  and  $v = x' + iy'$ , and regarding  $(x, y), (x', y')$  as the coordinates of the points  $P, P'$  in the two planes  $\Pi, \Pi'$  respectively, we have a real  $(m, n)$  correspondence between the two planes; viz. to each real point  $P$  in the first plane there correspond  $n$  real points  $P'$  in the second plane, and to each real point  $P'$  in the second plane there correspond  $m$  real points  $P$  in the first plane. But such real correspondence of two planes does not of necessity arise from an equation between the two imaginary variables  $u, v$ ; and the

question of the real correspondence of two planes may be considered in itself, without any reference to such origin.

I was under the impression that the theory was a known one; but I have not found it anywhere set out in detail. It is to be noticed that, although intimately connected with, it is quite distinct from (and seems to me to go beyond) that of a Riemann's surface. Riemann represents the value  $u, = x + iy$ , by a point  $P$  whose coordinates are  $x, y$ ; but he considers  $u', = x' + iy'$ , as a given imaginary value attached to the point  $P$ , without representing this value by a point  $P'$ , coordinates  $x', y'$ .

I proceed to consider the general theory of the real  $(m, n)$  correspondence. Points in the first plane are denoted by the unaccented letters  $P, Q, \dots$ ; and the corresponding points in the second plane are in general denoted by the same letters accented; but there are, as will be explained, special points  $V, W$ , where the points are interchanged; viz. to the points  $V$  or  $W$  in the first plane correspond  $W$  or  $V$  in the second plane.

1. To a point  $P$  there correspond in general  $n$  distinct points  $P'$ ; and as  $P$  varies continuously, each of the points  $P'$  also varies continuously.

2. There are certain points  $V$  called branch-points (*Verzweigungspunkte*), such that to each point  $V$  there correspond two united points, represented by  $(W)$ , and  $n - 2$  other distinct points  $W$ . The points  $(W)$  are called cross-points, and the number of them is of course equal to that of the branch-points  $V$ .

It is throughout assumed that a point denoted by a letter other than  $V$  is not a point  $V$ .

3. If the point  $P$ , moving continuously, describe a closed curve so as to return to its original position, then, if this curve includes within it no point  $V$  (or all the points  $V$ )<sup>1</sup>, each of the corresponding points  $P'$  will describe continuously a closed curve returning into its original position. Supposing that the curve described by  $P$  is an oval (non-autotomic closed curve), and taking this to be in the first instance an indefinitely small oval, then the curves described by the points  $P'$  will in the first instance be each of them an indefinitely small oval; but it is worth while to notice how, as the oval described by  $P$  increases, any one of the ovals described by a point  $P'$  may become autotomic; viz. if the oval described by  $P$  passes through two points  $Q, Q$  of the  $m$  points  $Q$  which correspond in the first plane to the same  $Q'$  in the second plane, then  $Q'$  will be a node in the closed curve described by that point  $P'$  which in the course of its motion comes to pass through  $Q'$ . This curve is in general an inloop curve composed of two loops, one wholly within the other (united at the point  $Q'$ ), and such that they each include one and the same point  $V$  (viz.  $V$  is included within the inner loop): as to this, see post, Nos. 9 and 10. It will be observed that this node  $Q'$  is not a point  $(W)$  nor any other special point of the second plane.

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<sup>1</sup>The two cases of the closed curve including no point  $V$ , and including all the points  $V$ , are really identical, as the discontinuity at infinity may be disregarded. It is to be observed that, this being so, it follows that the number of the points  $V$  must be even.

4. Consider, as before,  $P$  as describing a closed curve which does not include within it any point  $V$ , and the corresponding points  $P'$  as describing each of them a closed curve. As the curve described by  $P$  approaches a point  $V$ , the curves described by two of the points  $P'$  will approach the corresponding point ( $W$ ); and when the curve described by  $P$  passes through  $V$ , the curves described by the two points  $P'$  will unite together at this point ( $W$ ) as a node; viz. they will form a figure of eight<sup>2</sup>, the crossing being at the cross-point ( $W$ ), which corresponds to the branch-point  $V$ . And, corresponding to the closed curve described by  $P$ , we have this figure of eight (replacing two of the original  $n$  closed curves), and  $n - 2$  closed curves described by the other points  $P'$ .

5. Supposing, next, that the closed curve described by  $P$  (instead of passing through the point  $V$ ) includes within it the point  $V$ , then the figure of eight transforms itself into a twice-indented oval. There are on this curve two of the points  $P'$  which correspond to the given point  $P$ ; and as  $P$ , moving continuously in its closed curve, returns to its original position, the first of these points  $P'$ , moving continuously along a portion of the curve, comes to coincide with the original position of the second point  $P'$ ; while the second point  $P'$ , moving continuously along the remaining portion of the curve, comes to coincide with the original position of the first point  $P'$ ; viz. the two portions of the curve are described by the two points  $P'$  respectively. The curve may thus be regarded as a bifid curve, belonging to these two points  $P'$ . And, corresponding to the closed curve described by  $P$ , we have this bifid curve belonging to the two points  $P'$ , and  $n - 2$  single closed curves belonging to the other  $n - 2$  points  $P'$  respectively.

6. If the closed curve described by  $P$  (including within it a point  $V$ ) comes to pass through a second point  $V$ , the effect will be a new node at the corresponding point ( $W$ ); viz. at this point ( $W$ ) either the bifid curve unites itself with one of the single curves, or two of the single curves unite together, or the bifid curve there cuts itself. And, if the curve described by  $P$  comes to include within it this second point  $V$ , then in the three cases respectively: the bifid curve takes to itself the single curve, so that the system then is a trifold curve and  $n - 3$  single curves; or the two single curves give rise to a bifid curve, so that the system is two bifid curves and  $n - 4$  single curves; or, lastly, the bifid curve breaks up into two single curves, so that the system resumes its original form of  $n$  single curves.

7. We thus see how the closed curve described by  $P$ , including within it certain of the points  $V$ , may be such as to have corresponding to it an  $\alpha$ -fid curve, a  $\beta$ -fid curve, &c., ( $\alpha + \beta + \dots = n$ ); viz. an  $\alpha$ -fid curve contains upon it  $\alpha$  of the points  $P'$  which correspond to the original position of  $P$ ; and then, as  $P$  describes

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<sup>2</sup>The name *figure of eight* refers to the case where the two curves which come to unite at ( $W$ ) are proper ovals (non-autotomic closed curves). They might have one or both of them a node or nodes, as explained in No. 3; and the term would then be inappropriate. And so, lower down, the name twice-indented oval is used to express the form into which a proper figure of eight is changed by the disappearance of the node.

continuously its closed curve, returning to its original position, each of these points  $P'$  describes a portion of the  $\alpha$ -fid curve, passing from its original position to the original position of a point  $P'$  next to it upon the  $\alpha$ -fid curve; and the like as to a  $\beta$ -fid curve, &c. The numbers  $\alpha, \beta, \dots$  are not of necessity unequal, and we may have sets of equal numbers in any manner. It is hardly necessary to remark that, if the curve described by  $P$  passes through any point or points  $V$ , then two of the curves described by the points  $P'$  will unite together, or it may be that one of these will cut itself at the corresponding point or points ( $W$ ); and further that, as in No. 3, if the curve described by  $P$  passes through two or more of the points  $Q$  which correspond to the same point  $Q'$ , then any such point  $Q'$  will present itself as a node upon the curve belonging to some point, or set of points,  $P'$ . But the order of succession in which the original  $n$  single curves unite themselves together into multifid curves, or again break up into single curves, cannot, it would appear, be explained in any general manner, and would in each case depend on the nature of the particular correspondence.

8. We may consider the case where the closed curve described by  $P$  cuts itself. The curve may here be considered as made up of two or more ovals, or, to use a more appropriate term, say loops, each such loop being a curve not cutting itself; and the case is thus reducible to that before considered, where the curve does not cut itself. Thus, to fix the ideas, let the curve be a figure of eight, the initial position of  $P$  being at the crossing, and let neither of the loops contain within it a point  $V$ . Then, as  $P$  passes continuously along one of the loops, returning to its original position, each of the corresponding points  $P'$  describes a closed curve, which will be in the nature of a loop, viz. the initial and final directions of the motion of  $P$  not being continuous with each other, the initial and final directions of the motions of each point  $P'$  will not be continuous with each other, or there will be at the point  $P'$  an abrupt change in the direction of the curve. Similarly, as  $P$  describes the other loop of the figure of eight, each of the points  $P'$  will describe another loop; and the two loops belonging to the same point  $P'$  will unite together so as to form a figure of eight; viz. to the figure of eight described by  $P$  there will correspond figures of eight described by the  $n$  points  $P'$  respectively.

9. But consider next the case where the two loops of the curve described by  $P$  include each of them one and the same point  $V$ . This implies that one of the two loops lies inside the other, or that the curve is what has been called an inloop curve. As  $P$ , which is in the first instance taken to be at the node, passes continuously along one of the loops and returns to its original position, there are two of the points  $P'$  such that the first of these passes from its original position to the original position of the second, and the second of them passes from its original position to the original position of the first of them. We have thus two arcs between these two points  $P'$ ; but inasmuch as the initial and the final directions of motion of the point  $P$  are not continuous with each other, these two arcs are not continuous in direction at the two points  $P'$ , but at each of these points  $P'$  the two arcs meet at an angle. As  $P$  describes the other loop, we have in like manner two arcs between the same two points  $P'$ , these arcs at each of the points  $P'$  meeting at an

angle; but they join on to the first-mentioned two arcs in such manner as to form two ovals intersecting each other in the two points  $P'$ . Corresponding to the inloop curve described by  $P$ , we have this pair of intersecting ovals described by two of the points  $P'$ , and  $n - 2$  other curves described by the other points  $P'$ , and being each of them (I assume) an inloop curve.

10. If we attend only to one of the two intersecting ovals, we have in the first plane an inloop curve, and corresponding thereto in the second plane an oval passing through two of the points  $P'$  which correspond to the node  $P$  of the inloop curve. Interchanging the two planes, and writing  $Q$  instead of  $P$ , we have in the first plane an oval passing through two of the points  $Q$  which correspond to a point  $Q'$ ; and corresponding to this oval we have in the second plane an inloop curve having this point  $Q'$  for its node, viz. these are the corresponding figures mentioned in No. 3.

11. Consider a given point  $Q$ ; and let the corresponding points  $Q'$  be called (selecting the suffixes at pleasure)  $Q'_1, Q'_2, \dots, Q'_n$ . Taking then a point  $O$  indefinitely near to  $Q$ , the corresponding points  $O'$  will be indefinitely near to  $Q'_1, Q'_2, \dots, Q'_n$  respectively, and they will be called  $O'_1, O'_2, \dots, O'_n$  accordingly. It is to be observed that by the indefinitely near point  $O$  is meant a point such that the distance from  $O$  to  $Q$  is indefinitely small in comparison with the distance of either of these points from any point  $V$ ; so that we cannot have from  $Q$  to  $O$  two indefinitely short paths including between them a point  $V$ ; or say so that the indefinitely short path from  $Q$  to  $O$  is determinate.

Proceeding in this manner from  $Q$  to  $O$ , and so through a succession of indefinitely near points to a distant point  $S$ , we seem to determine the suffixes of the corresponding points  $S'$ ; but, by what precedes, it appears that such determination for a point  $S$  is dependent on the path from  $Q$  to  $S$ ; and consequently that we do not thus obtain a proper determination of the suffixes of the points  $S'$ . In fact, if we were to pass from  $Q$  by a path including one or more of the points  $V$  back to  $Q$ , we should obtain for the several points  $Q'$  respectively suffixes which are in general different from the suffixes originally given to these points respectively.

12. The difficulty is got over as follows:—Considering as before the given point  $Q$ , and calling the corresponding points  $Q'_1, Q'_2, \dots, Q'_n$  at pleasure, we pass from  $Q$  to the indefinitely near point  $O$ , and thence, by so many paths chosen at pleasure, to the several branch-points  $V$ ; these paths from  $O$  to the several points  $V$  are called barriers. To fix the ideas, we may consider these as non-autotomic non-intersecting lines drawn from  $O$  to the several points  $V$ . Consider the barrier from  $O$  to one of these points  $V$ ; as  $P$  passes along this barrier from  $O$  to  $V$ , two of the corresponding points  $P'$  will pass from two of the corresponding points  $O'$  to the corresponding cross-point ( $W$ ); the paths of these two points are called the counter-barrier corresponding to the barrier in question; and we have thus in the second plane a system of counter-barriers, each drawn from two points  $O'$  to meet in a point ( $W$ ). By what precedes, the points  $O'$  have each of them a determinate suffix; a counter-barrier is thus drawn from two points with given suffixes, suppose  $O'_1$  and  $O'_2$ , to a



point ( $W$ ), and this may be distinguished accordingly as a counter-barrier 12; and in like manner the cross-point ( $W$ ) through which it passes will be called a cross-point ( $W_{12}$ ); and the barrier corresponding hereto, and the branch-point  $V$  at which it terminates, will in like manner be called a barrier 12, and a branch-point  $V_{12}$ . Each barrier and branch-point will thus have a pair of suffixes; and the corresponding counter-barrier and cross-point will have the same pair of suffixes. It is to be observed that two or more of these corresponding figures may very well have the same pair of suffixes; but that such corresponding figures must be distinguished from each other; thus, if there are two branch-points  $V_{12}$ , these may be distinguished as the branch-points  $\alpha V_{12}$  and  $\beta V_{12}$ , and the barriers, counter-barriers, and cross-points by means of these same letters  $\alpha$  and  $\beta$ , (or otherwise), as may be convenient. It would seem that not only the number of the points  $V$  must be even, but the number of each set of points  $V_{12}$  must also be even (see post, No. 15).

13. It is also to be noticed that the determination of the suffixes of the several points  $V$ , &c., depends first upon the arbitrary choice of the suffixes of the points  $Q'$ , and next on the choice of the system of barriers; but that, these being assumed, the suffixes of the several points  $V$ , &c., are completely determinate.

14. Taking now any point  $S$  whatever, and supposing that  $P$  moves from  $Q$  continuously to  $S$  by a path which does not meet a barrier, the points  $P'$  will move from the several points  $Q'$  to the several points  $S'$  by paths not meeting the counter-barriers; viz. to each point  $S'$  there will be a path from some point  $Q'$ ; and giving to such point  $S'$  the suffix of the point  $Q'$ , the suffixes of the several points  $S'$  which correspond to any point whatever,  $S$ , will be completely determined. The determination depends of course on the assumptions referred to No. 13, but not in anywise on the position of the point  $S$ .

It will be noticed that, as all the points  $V$  are connected together by the barriers, the only closed paths from a point to itself are paths not including any, or including all, of the points  $V$ ; and that between such paths there is no real distinction.

15. Consider a point  $P$  moving continuously in any manner. The several corresponding points  $P'_1, P'_2, \dots, P'_n$  will each of them move continuously, but the suffixes interchange; viz. when  $P$  arrives at and then passes over a barrier  $\alpha\beta$ , the corresponding points  $P'_\alpha$  and  $P'_\beta$  will each arrive at the corresponding counter-barrier  $\alpha\beta$ , and, on passing over this,  $P'_\alpha$  will be changed into  $P'_\beta$  and  $P'_\beta$  into  $P'_\alpha$ , the other points  $P'$  remaining unchanged; and the like in other cases. This in fact includes the whole or the greater part of the foregoing theory. Thus, if  $P$  describe a closed curve not cutting any barrier, there will be no change of suffix; and when  $P$  returns to its original position each of the corresponding points  $P'_1, P'_2, \dots, P'_n$  will describe a closed curve, returning to its original position. But suppose that  $P$  describes a closed curve, cutting once only a barrier 12; suppose that the path is from  $P$  to  $Q$ , and then crossing the barrier to  $R$ , and thence again to  $P$ ;  $P'_1$  passes to  $Q'_1$ , and then crossing the counter-barrier it passes from  $P'_1$  to  $P'_2$ ; while at the same time  $P'_2$  passes to  $Q'_2$ , and then crossing the counter-barrier it passes from  $R'_2$  to  $P'_1$ ;

viz. we have  $P'_1, P'_2$  describing the two portions of a bifid curve. If there were only a single branch-point  $V_{12}$ , and therefore only a single barrier  $OV_{12}$ , then we might have through  $P$  a closed curve cutting  $OV_{12}$  once only, and including within it the point  $O$ , but not including within it the point  $V_{12}$ ; and here there ought not to be a bifid curve, but the points  $P'_1, P'_2$  ought to describe each of them a single curve. But suppose there are two points  $V_{12}$ , and consequently two barriers  $OV_{12}$  (meeting in  $O$ ); then the closed curve, meeting once only a barrier 12, (viz. it meets only one such barrier, and that once only), must include within one and only one of the two points  $V_{12}$ ; and in this case there ought to be a bifid curve. It is by such reasoning as this that I infer the foregoing theorem (No. 12), that the number of each set of points  $V_{12}$  is even.

16. We may consider how the suffixes are affected when, instead of the original system of barriers, we have a new system of barriers. I suppose that we have in the two cases respectively the same point  $Q$ , and the same suffixes for the points  $Q'_1, Q'_2, \dots, Q'_n$  which correspond thereto. In the first case, passing from  $Q$  to an indefinitely near point  $O$ , say the red  $O$ , we draw from this point to the several points  $V$  a set of barriers, say the red barriers; while in the second case, passing from  $Q$  to an indefinitely near point  $O$ , say the blue  $O$ , we draw from this point to the several points  $V$  a set of barriers, say the blue barriers; and we then proceed as before, viz. in the first case, drawing from  $Q$  to the point  $S$  a curve which does not meet any of the red barriers, we determine accordingly the suffixes (say the red suffixes) of the several corresponding points  $S'$ ; and in the second case, drawing in like manner from  $Q$  to  $S$  a curve which does not meet any of the blue barriers, we determine accordingly the suffixes (say the blue suffixes) of the same points  $S'$ . Now the curve drawn from  $Q$  to  $S$  so as not to cut any of the red barriers, and which is used for the determination of the red suffixes of the several points  $S'$ , will in general cut certain of the blue barriers; and, by examining the suffixes of the blue barriers which are thus cut, we determine the blue suffixes of the same points  $S'$ ; the result of course depending only on the situation of  $S$  in one or other of the regions formed by the red barriers and the blue barriers conjointly. In particular, the point  $S$  may be so situate that we can from  $Q$  to  $S$  draw a curve not meeting any red barrier or any blue barrier; and in this case the red suffixes and the blue suffixes are identical.

17. We may imagine the first plane as consisting of  $n$  superimposed planes or sheets, say the sheets  $1, 2, \dots, n$ . Each barrier 12 is considered as a line drawn in the two sheets 1 and 2; and so on in other cases. The point  $P$  is considered as a set of superimposed points  $P_1, P_2, \dots, P_n$  moving in the several sheets respectively; under the convention that  $P_1$ , moving in the sheet 1, and coming to cross a barrier 12, passes into the sheet 2 and becomes  $P_2$ ; and the like in other cases. And this being so, we say that to a point  $P$ , considered as a point  $P_\alpha$  in the sheet  $\alpha$ , there corresponds in the second plane one and only one point  $P'_\alpha$ ; and that  $P$  moving continuously in any manner (subject to the change of sheet as just explained), each of the  $n$  corresponding points  $P'$  will also move continuously, and so that each such point  $P'_\alpha$  will return

to its original position, upon the corresponding point  $P_\alpha$  returning to its original position and sheet. This is, in fact, Riemann's theory, only instead of the points  $P'$  we must speak of the values  $x' + iy'$  of the irrational function of  $x + iy$ .

18. Everything is of course symmetrical as regards the two planes; we have therefore, in the second plane, a system of points  $V$  and of barriers, and in the first plane a system of points  $(W)$  and of counter-barriers. To a given point  $P'$  in the second plane there correspond  $m$  points  $P$  in the first plane; and we can (the determination depending on the system of barriers in the second plane) assign to the  $m$  points suffixes, thereby distinguishing them as the corresponding points  $P_1, P_2, \dots, P_m$ . And we may imagine the second plane as consisting of  $m$  superimposed planes or sheets, say the sheets  $1, 2, 3, \dots, m$ ; the general theorem then is that to a point  $P$  or  $P'$  in either plane, considered as a point  $P_\alpha$  or  $P'_{\alpha'}$  in the sheet  $\alpha$  or  $\alpha'$ , there corresponds in the other plane one and only one point  $P'_{\alpha'}$  or  $P_\alpha$ ; and that the first-mentioned point in either plane moving continuously in any manner (subject to the proper change of sheet), the corresponding point in the other plane will also move continuously, and will return to its original position and sheet, upon the first-mentioned point returning to its original position and sheet.

19. In all that precedes it has been assumed that, to a branch-point  $V$ , there correspond two united points represented by  $(W)$  and  $n-2$  distinct points  $W$ ; the cases of a point  $(W)$  composed of three or more united points, or of the points  $W$  uniting themselves in sets in any other manner, would give rise to further specialities.

## 690.

## ON THE THEORY OF GROUPS.

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I recapitulate the general theory so far as is necessary in order to render intelligible the quasi-geometrical representation of it which will be given.

Let  $\alpha, \beta, \dots$  be functional symbols each operating upon one and the same number of letters, and producing as its result the same number of functions of these letters. For instance,  $\alpha(x, y, z) = (X, Y, Z)$ , where the capitals denote each of them a given function of  $(x, y, z)$ .

Such symbols are susceptible of repetition and combination

$$\alpha^2(x, y, z) = \alpha(X, Y, Z),$$

or

$$\beta\alpha(x, y, z) = \beta(X, Y, Z),$$

in each case equal to three given functions of  $(x, y, z)$ ; and similarly for  $\alpha^3, \alpha^2\beta$ , etc.

The symbols are not in general commutative,  $\alpha\beta$  not  $= \beta\alpha$ ; but they are associative,  $\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma$ , each  $= \alpha\beta\gamma$ , which has thus a determinate meaning.

Unity as a functional symbol denotes that the letters are unaltered,

$$1(x, y, z) = (x, y, z);$$

whence

$$1\alpha = \alpha 1 = \alpha.$$

The functional symbols may be substitutions;  $\alpha(x, y, z) = (y, z, x)$ , the same letters in a different order. Substitutions can be represented by the notation  $\begin{pmatrix} x & y & z \\ y & z & x \end{pmatrix}$ , the substitution which changes  $xyz$  into  $yzx$ , or, as products of cyclical substitutions,  $\begin{pmatrix} x & y & z & u & v & w \\ y & z & x & w & v & u \end{pmatrix} = (xyz)(uw)$ , the product of the cyclical substitutions  $x$  into  $y$ ,  $y$  into  $z$ ,  $z$  into  $x$ , and  $u$  into  $w$ ,  $w$  into  $u$ , the letter  $v$  being unaltered.

A set of symbols  $\alpha, \beta, \gamma, \dots$ , such that the product  $\alpha\beta$  of each two of them (in each order,  $\alpha\beta$  and  $\beta\alpha$ ) is a symbol of the set, is a group. It is easily seen that 1 is a symbol of every group, and we may therefore give the definition in the form that a set of symbols  $1, \alpha, \beta, \gamma, \dots$  satisfying the foregoing condition is a group. When the number of symbols (or terms) is  $= n$ , then the group is of the order  $n$ ; and each symbol  $\alpha$  is such that  $\alpha^n = 1$ , so that a group of the order  $n$  is in fact a group of symbolical  $n$ th roots of unity.

A group is defined by means of the laws of combinations of its symbols. For the statement of these we may either (by the introduction of powers and products) diminish as much as may be the number of distinct functional symbols; or else, using distinct letters for the several terms of the group, employ a square diagram, as presently mentioned.

Thus, in the first mode, a group is  $1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2$ , ( $\alpha^2 = 1, \beta^3 = 1, \alpha\beta^2 = \beta\alpha$ ), where observe that these conditions imply also  $\alpha\beta^2 = \beta\alpha$ .

Or in the second mode, calling the symbols  $(1, \alpha, \beta, \alpha\beta, \beta^2, \alpha\beta^2)$  of the same group  $(1, \alpha, \beta, \gamma, \delta, \epsilon)$ , or, if we please,  $(a, b, c, d, e, f)$ , the laws of combination are given by one or other of the square diagrams

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$      | $\epsilon$ |
|------------|------------|------------|------------|------------|---------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$      | $\epsilon$ |
| $\alpha$   | $\alpha$   | 1          | $\gamma$   | $\beta$    | $\epsilon$    | $\delta$   |
| $\beta$    | $\beta$    | $\epsilon$ | $\delta$   | $\alpha$   | 1             | $\gamma$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha\beta$ |            |
| $\delta$   | $\delta$   | $\gamma$   | 1          | $\epsilon$ | $\beta$       | $\alpha$   |
| $\epsilon$ | $\epsilon$ | $\beta$    | $\alpha$   | $\delta$   | $\gamma$      | 1          |

| $f$ | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|-----|-----|-----|-----|-----|-----|-----|
| $f$ | $b$ | $a$ | $d$ | $c$ | $f$ | $e$ |
| $f$ | $c$ | $e$ | $b$ | $a$ | $d$ |     |
| $f$ | $d$ | $e$ | $a$ | $b$ | $c$ |     |
| $f$ | $e$ | $d$ | $a$ | $c$ | $b$ |     |
| $f$ | $c$ | $b$ | $e$ | $d$ | $a$ |     |

where, taking for greater symmetry the second form of the square, observe that the square is such that no letter occurs twice in the same line, or in the same column (or what is the same thing, each of the lines and of the columns contains all the letters). But this is not sufficient in order that the square may represent a group; the square must be such that the substitutions by means of which its several lines are derived from any line thereof are (in a different order) the same substitutions by which the lines are derived from a particular line, or say from the top line. These, in fact, are

$$\begin{aligned}
 &ab \cdot cd \cdot ef, \\
 &ace \cdot bfd, \\
 &ad \cdot be \cdot cf, \\
 &aec \cdot bdf, \\
 &af \cdot be \cdot de,
 \end{aligned}$$

where, for shortness,  $ab$ ,  $ace$ , &c., are written instead of  $(ab)$ ,  $(ace)$ , &c., to denote the cyclical substitutions  $a$  into  $b$ ,  $b$  into  $a$ ; and  $a$  into  $c$ ,  $c$  into  $e$ ,  $e$  into  $a$ , &c.; and it is at once seen that by the same substitutions the lines may be derived from any other line.

It will be noticed that in the foregoing substitution-group each substitution is regular, that is, composed of cyclical substitutions each of the same number of letters; and it is easy to see that this property is a general one; each substitution of the substitution-group must be regular.

By what precedes, the group of any order composed of the functional symbols is replaced by a substitution-group upon a set of letters the number of which is equal to the order of the group, and wherein all the substitutions are regular.

The general theory being thus explained, I endeavour to form a substitution-group with the twelve letters  $abcdefghijkl$ ; and I assume that there is one substitution, such as  $abc \cdot def \cdot ghi \cdot jkl$ , and another substitution, such as  $agj \cdot bfi \cdot cek \cdot dhl$ . Observe that, if the twelve letters are to be thus arranged in two different ways as a set of four triads, without repetition of any duad, all the ways in which this can be done are essentially similar, and there is no loss of generality in taking the two sets of triads to be those just written down. But the substitution to be formed with either set of triads will be different according as any triad thereof, for instance  $agj$ , is written in this form or in the reversed form  $ajg$ . There are thus in all sixteen substitutions which can be formed with the first set of triads, and sixteen substitutions which can be formed with the second set of triads; and the relation of a triad of the first set to a triad of the second set is by no means independent of the selection of the triads out of the two sets respectively. To show this, take the two substitutions quite at random; suppose they are those written down above, say

$$\alpha = abc \cdot def \cdot ghi \cdot jkl, \quad \beta = agj \cdot bfi \cdot cek \cdot dhl$$

and perform these in succession on the primitive arrangement  $\Omega = abcdefghijkl$ . The operation stands thus:

$$\beta\alpha\Omega = fegkihbjcda,$$

$$\alpha\Omega = bcaefdhigklj,$$

whence

$$\Omega = abcdefghijkl,$$

$$\beta\alpha = abeijcgl \cdot dk,$$

is not a regular substitution; and, by what precedes,  $\alpha, \beta$  cannot belong to a group.

But take the substitutions to be

$$\alpha \text{ (as before)} = abc \cdot def \cdot ghi \cdot jkl, \quad \beta = ajg \cdot bif \cdot cek \cdot dhl,$$

then we have

$$\beta\alpha\Omega = iejkbhlfacdg,$$

$$\alpha\Omega = bcaefdhigklj,$$

$$\Omega = abcdefghijkl,$$

whence

$$\beta\alpha = ai \cdot bc \cdot cj \cdot dk \cdot fh \cdot gl,$$

a regular substitution; and, for anything that appears to the contrary,  $\alpha, \beta$  may belong to a group. It is convenient to mention at once that these two substitutions do, in fact, give rise to a group; viz. the square diagram is

|          |          |          |          |          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>e</i> | <i>f</i> | <i>g</i> | <i>h</i> | <i>i</i> | <i>j</i> | <i>k</i> | <i>l</i> |
| <i>b</i> | <i>c</i> | <i>a</i> | <i>f</i> | <i>d</i> | <i>e</i> | <i>h</i> | <i>i</i> | <i>g</i> | <i>k</i> | <i>l</i> | <i>j</i> |
| <i>c</i> | <i>a</i> | <i>b</i> | <i>e</i> | <i>f</i> | <i>d</i> | <i>i</i> | <i>g</i> | <i>h</i> | <i>l</i> | <i>j</i> | <i>k</i> |
| <i>d</i> | <i>l</i> | <i>h</i> | <i>g</i> | <i>j</i> | <i>c</i> | <i>e</i> | <i>k</i> | <i>f</i> | <i>i</i> | <i>b</i> | <i>a</i> |
| <i>e</i> | <i>j</i> | <i>i</i> | <i>b</i> | <i>h</i> | <i>k</i> | <i>a</i> | <i>l</i> | <i>d</i> | <i>g</i> | <i>c</i> | <i>f</i> |
| <i>f</i> | <i>k</i> | <i>g</i> | <i>c</i> | <i>i</i> | <i>l</i> | <i>b</i> | <i>j</i> | <i>e</i> | <i>h</i> | <i>a</i> | <i>d</i> |
| <i>g</i> | <i>f</i> | <i>k</i> | <i>l</i> | <i>e</i> | <i>i</i> | <i>j</i> | <i>a</i> | <i>c</i> | <i>b</i> | <i>d</i> | <i>h</i> |
| <i>h</i> | <i>d</i> | <i>l</i> | <i>j</i> | <i>a</i> | <i>g</i> | <i>k</i> | <i>c</i> | <i>b</i> | <i>f</i> | <i>i</i> | <i>e</i> |
| <i>i</i> | <i>e</i> | <i>j</i> | <i>k</i> | <i>b</i> | <i>h</i> | <i>l</i> | <i>f</i> | <i>a</i> | <i>c</i> | <i>d</i> | <i>g</i> |
| <i>j</i> | <i>i</i> | <i>e</i> | <i>h</i> | <i>k</i> | <i>b</i> | <i>a</i> | <i>d</i> | <i>l</i> | <i>f</i> | <i>g</i> | <i>c</i> |
| <i>k</i> | <i>g</i> | <i>f</i> | <i>i</i> | <i>l</i> | <i>c</i> | <i>b</i> | <i>j</i> | <i>d</i> | <i>h</i> | <i>a</i> | <i>e</i> |
| <i>l</i> | <i>h</i> | <i>d</i> | <i>g</i> | <i>j</i> | <i>a</i> | <i>f</i> | <i>k</i> | <i>e</i> | <i>i</i> | <i>b</i> | <i>c</i> |

and the substitutions, obtained therefrom by writing successively each line over the top line, are

$$\begin{aligned}
 1 &= 1, \\
 abc \cdot def \cdot ghi \cdot jkl &= \alpha, \\
 acb \cdot dfe \cdot gih \cdot jlk &= \alpha^2, \\
 ad \cdot bl \cdot ch \cdot eg \cdot fj \cdot ik &= \beta\alpha^2, \\
 aeh \cdot bjd \cdot cli \cdot fkg &= \beta\alpha^2, \\
 afl \cdot bkh \cdot cgd \cdot eij &= \beta^2\alpha, \\
 agj \cdot bfi \cdot cek \cdot dlh &= \beta\alpha, \\
 ahe \cdot bdj \cdot cli \cdot fgk &= \beta\alpha^2\beta\alpha^2, \\
 ai \cdot bc \cdot cj \cdot dk \cdot fh \cdot gl &= \beta\alpha, \\
 ajg \cdot bif \cdot cek \cdot dhl &= \beta, \\
 ak \cdot bg \cdot cf \cdot di \cdot el \cdot hj &= \beta\alpha\beta, \\
 alf \cdot bkh \cdot cdg \cdot eji &= \beta^2\alpha^2\beta\alpha.
 \end{aligned}$$

To explain the theory, I introduce the notion of a hemipolyhedron, or say a hemihedron, viz. this is a figure obtained from a polyhedron by the removal of certain faces. In a polyhedron each edge occurs twice (more properly it occurs in the two forms  $ab$  and  $ba$ ), as belonging to two faces; but in a hemihedron one of these faces must always be removed, so that the edge may occur once only; and again (what is apparently, although not really, a different thing), we may remove two intersecting faces, leaving their edge of intersection; this edge is, in fact, then considered as a bilateral face  $ab = ab \cdot ba$ , just as  $abc$  is a trilateral face  $abc = ab \cdot bc \cdot ca$ . Thus, if in a prism we remove the lateral faces, leaving the lateral edges, and leaving also the terminal faces, we have a hemihedron: thus, the prism being trilateral, say the faces of the hemihedron are  $abc, def, ad, be, cf$ , where  $ad, be, cf$  are the edges regarded as bilateral faces. And, for the present purpose,  $abc$  denotes the cyclical substitution  $a$  into  $b$ ,  $b$  into  $c$ ,  $c$  into  $a$ ; and  $ad$  denotes in like manner the cyclical substitution (or interchange)  $a$  into  $d$ ,  $d$  into  $a$ .

But the hemihedron about to be considered has no bilateral faces; it is, in fact, the figure composed of the 8 triangular faces of the octo-hexahedron or figure obtained by truncating the summits of a hexahedron (or of an octahedron) so as to obtain a polyhedron of 8 triangular faces and 6 square faces, representing the faces of the octahedron and the hexahedron respectively. The faces of the octo-hexahedron may be taken to be

$$\begin{aligned} abc, def, ghi, jkl, \\ agj, bfi, cek, dhl, \\ cbfe, fihd, hgjl, jack, agib, klde, \end{aligned}$$

(where I observe in passing that the symbols are written in such manner that each edge  $ab$  occurs under the two opposite forms  $ab$  in  $abc$  and  $ba$  in  $agib$ ). And then, omitting the square faces, represented by the third line, we have the hemihedron, wherein as before  $abc$  denotes the cyclical substitution  $a$  into  $b$ ,  $b$  into  $c$ ,  $c$  into  $a$ ; and so for the other faces.

I represent this by a diagram, the lines of which were red and black, and they will be thus spoken of, but the black lines are in the woodcut continuous lines, and the red lines broken lines: each face indicates a cyclical substitution, as shown by the arrows. The figure should be in the first instance drawn with the arrows, but without the letters, and these may then be affixed to the several points in a perfectly arbitrary manner; but I have in fact affixed them in such wise that the group given



by the diagram, as presently appearing, may (instead of being any other equivalent group) be that group which contains the before-mentioned substitution

$$\alpha = abc \cdot def \cdot ghi \cdot jkl, \quad \text{and} \quad \beta = ajg \cdot bif \cdot cek \cdot dhl.$$

Observe that in the diagram, considering the lines to be drawn as shown by the arrows, there is from any given point whatever only one black line, and only one red line. Let  $B$  denote motion along a black line,  $R$  motion along a red line (always from a point to the next point); then  $B^2$  will denote motion along two black lines successively,  $RB$  (any such symbol being read always from right to left) will denote motion first along a red line, and then along a black line, and so in other cases; a symbol or "route"  $\dots R^\rho B^\beta R^{\rho'}$  has thus a perfectly definite signification, determining the path when the initial point is given.

The diagram has the property that every route, leading from any one letter to itself, leads also from every other letter to itself; or say a route leading from  $a$  to  $a$ , leads also from  $b$  to  $b$ , from  $c$  to  $c$ ,  $\dots$ , from  $l$  to  $l$ ; and we can thus in the diagram speak absolutely (that is, without restriction as to the initial point) of a route as leading from a point to itself, or say as being equal to unity; it is in virtue of this property that the diagram gives a group.

For, assuming the property, it at once follows (1) that two routes, each leading say from the point  $a$  to the same point  $f$ , lead also from any other point  $b$  to one and the same point  $g$ . Such routes are said to be equivalent, or equal to each other; and the number of distinct routes (including the route unity) is thus equal to the numbers of the letters, viz. we have only the routes from  $a$  to  $a$ , to  $b$ ,  $\dots$ , to  $l$ , respectively; (2) a route, leading from a point  $a$  to a point  $f$ , leads from any other point  $b$  to a different point  $g$ ; and (3) two routes, leading from the same point  $a$  to different points  $b$  and  $c$ , lead also from any other point  $f$  to different points  $k$  and  $l$ . Hence a given route leads from the several points  $abc \dots l$  successively to the same series of points taken in a different order, or we thus obtain a new arrangement of the points; and dealing in this manner successively with the routes from  $a$  to  $a$ , to  $b$ ,  $\dots$ , to  $l$ , we obtain so many distinct arrangements, beginning with the letters  $a, b, c, \dots, l$  respectively, such that in no two of them does the same letter occupy the same place; we thus obtain a square of 12 such as that already written down, and which is, in fact, the same square, the several routes of course corresponding to the substitutions of the square. The hemihedron thus gives the foregoing group of 12.

Observe that the diagram is composed of the four black triangles representing the substitution  $abc \cdot def \cdot ghi \cdot jkl$ , and of the four red triangles representing the substitution  $ajg \cdot bif \cdot cek \cdot dhl$ ; viz. these are independent substitutions which by their powers and products serve to express all the substitutions of the group; that they are sufficient appears by the diagram itself, in that every point thereof is (by black and red lines) connected with every other point thereof. The group might have contained three or more independent substitutions, and the diagram would then have contained the like number of differently coloured sets of lines. The essential characters are that the lines of any given colour shall form polygons of the same number of sides (but for different

colours the polygons may have different numbers of sides; in particular, for any given colour or colours, the polygons may be bilaterals, represented each by a line with a double arrow pointing opposite ways); that there shall be from each point only one line of the same colour; that every point shall be connected with every other point; and finally, that every route leading from one point to itself shall lead also from every other point to itself. When these conditions are satisfied the foregoing investigation in fact shows that the diagram, or say the hemihedron, gives rise to a group.

It may be remarked that we can, if we please, introduce into the diagram a set of lines of a new colour to represent any dependent substitution of the group; thus, in the example considered, a substitution is  $ae h \cdot b j d \cdot c i l \cdot f k g$ , and if we draw these triangles in green (the arrows being from  $a$  to  $e$ ,  $e$  to  $h$ ,  $h$  to  $a$ , &c.), then there will be from each point one black line, one red line, and one green line; any route  $\dots G^\gamma B^\beta R^\rho$  will thus be perfectly definite, and will have the same properties as a route composed of black and red lines only; and the theory thus subsists without alteration.

I remark, in conclusion, that the group of 12 considered above is, in fact, the group of 12 positive substitutions upon 4 letters  $abcd$ ; viz. the substitutions are 1,  $abc$ ,  $acb$ ,  $abd$ ,  $adb$ ,  $acd$ ,  $adc$ ,  $bcd$ ,  $bdc$ ,  $ab \cdot cd$ ,  $ac \cdot bd$ ,  $ad \cdot bc$ ; the groups each contain unity, three substitutions of the order (or index) 2, and 8 substitutions of the order (or index) 3, and their identity can be easily verified.